# The Household Demand for Leisure, the Price of Time and the Full Cost of Children: A Structural Model and Evidence from the PSID*

Olivier Donni Anderson Vil

September 18, 2026

**Abstract**

We propose a novel approach to estimating the full cost of children — the sum of monetary and time costs — by endogenizing the price of parental time rather than assuming it is equal to the parents' wage rate. In this approach, the price of time depends on how parents perceive their time with children, whether as a leisure-like activity or more as a labor-like activity. We then develop a simplified collective model of leisure demand for working couples, incorporating individual preferences and childcare technology, and estimate it using 2019 PSID data. This allows us to recover the price of parental time and the full cost of children. We find that mothers perceive 44% of their childcare time as labor, compared to 35% for fathers. Our results also highlight that a substantial portion of the full cost of children is non-monetary.

JEL Classification: I30, I31, J21, J22, J31

Keywords: Labor Supply, Leisure Demand, Collective Model, Price of time, Identification, Resource sharing, Cost of children.

*Affiliations: Olivier Donni — THEMA, CY Cergy Paris Université; Anderson Vil — Paris School of Economics. We thank Olivier Bargain, Laurens Cherchye, Marion Goussé, Elena Stancanelli, and Alexandros Theloudis for helpful comments. Financial support from ANR WHIN (ANR-21-ESRE-0034) is gratefully acknowledged.

# 1 Introduction

The cost of children is a critical parameter used to implement many economic policies or to calculate inequality measures. Economists have long developed methods to infer from survey data what parents spend for children (see Bargain and Donni, 2012; Bargain, Donni and Hentati, 2022; Bargain, Lacroix, and Tiberti, 2022; Brown, Calvi, and Penglase, 2021; Calvi et al., 2023; Dunbar, Lewbel and Pendakur, 2013; Lechene, Pendakur, and Wolf, 2022; Lewbel and Pendakur, 2024; Penglase, 2021, for recent applications). However, focusing solely on monetary expenditures provides an incomplete picture of the full cost of children. The full cost of children indeed consists of both a monetary cost — the purchase of goods and services that contribute to children's well-being — and a time cost, which reflects the value of time parents and other caregivers dedicate to children.[1]

Evaluating the time cost is particularly complex and has been largely overlooked, with only a few exceptions. Gustafsson and Kjulin (1994) value parents' time by using either wage rates or the price of equivalent services. They then combine this with survey-based measures of monetary expenditures to construct a measure of the full cost of children. Apps and Rees (2001) develop a structural model to estimate the full cost of children, incorporating both monetary and time costs. Their model assumes linear homogeneity in childcare technology, identical parental preferences, and that the price of time is determined by the parents' wage rates. Similarly, Colombino (2000) constructs a simple structural model where the number of children is endogenous and parental time is also valued at the wage rate. Finally, Bradbury (2008) presents a theoretical framework and numerical illustrations, where the full cost of children is inferred from variations in parents' leisure, following a Rothbarth-style approach.[2]

One of the main challenges in evaluating the full cost of children is assigning a value to childcare

[1] To be comprehensive, other long-term costs may also arise, such as reduced career advancement opportunities (see Adda, Dustmann, and Stevens, 2017; Budig and England 2001; Dechter, 2014; Glauber, 2018; Korenman and Neumark, 1992; Waldfogel, 1998).

[2] Koulovatianos, Schröder and Schmidt (2009) adopt a different identification strategy, recovering the full cost of children from subjective questions rather than observed parental behavior.

time. Typically, the price of time a spouse dedicates to any productive activity is assumed to align with their wage rate. However, this approach relies on two strong hypotheses: (a) the spouse must participate in the labor market and be able to adjust working time sufficiently freely for the observed wage to provide a meaningful measure of the price of time, and (b) the time devoted to the activity must be perfectly substitutable for market work, with a marginal rate of substitution equal to one. If these conditions do not hold, the value assigned to parental childcare time in the measurement of the time cost of children must instead be determined endogenously from parental preferences and the childcare technology. The first condition is a general difficulty in valuing household production and depends largely on labor-market institutions and constraints. The second is particularly problematic in the childcare context. For example, Cosaert and Hennebel (2023) show that a large fraction of childcare time is perceived by parents as leisure, suggesting that it should not be valued entirely as a cost for them. This idea is formalized by Juster and Stafford (1991) through the concept of 'process benefits', which captures how certain activities generate direct utility for individuals beyond their intended productive outcome. Reid's (1934) third-party criterion provides a way of distinguishing household production from pure consumption: an activity is productive when it is undertaken to generate an outcome that could, in principle, be provided by a third party. Together, these two concepts suggest that childcare time may simultaneously have a productive dimension and generate direct utility for parents, while providing a basis for distinguishing the productive value of this time from the process benefits associated with performing the activity.

The aim of this paper is to evaluate the full cost associated with raising children, using a structural framework that identifies the productive component of parental childcare time. To achieve this, we develop a collective labor supply model for a working couple with children. Each parent has a distinct utility function that depends on his or her leisure and childcare time, own consumption, and children's consumption, with the crucial feature that childcare time is not necessarily equivalent to market work from the parent's perspective. More generally, childcare time can be viewed by parents as either a leisure-like activity that increases utility — reflecting positive process benefits — or a labor-like activity that has no direct effect on utility or may

even reduce it, corresponding to negative process benefits.[3] In addition to influencing utility, childcare time also enters a childcare technology together with purchased childcare services. Inspired by Reid's (1934) third-party criterion, this technology does not represent the production of childcare itself, but rather the substitution possibilities between parental childcare time and market-provided childcare services. The price of parental childcare time is determined by the marginal rate of substitution between these two inputs, while its productive value reflects the extent to which it substitutes for market-provided childcare. To complete the model, we finally assume that parents incur expenses for their children according to a predetermined rule. The full cost of children is finally assumed to comprise both the value of parental time devoted to the productive childcare activity and the direct expenses for children.

Our main result is that the full cost of children can be identified from observed behavior. Specifically, our model allows us to break down each unit of childcare time into a productive childcare activity and a leisure-like (or labor-like) activity that directly affects the parents' utility. We then estimate this model using a sample of dual-earner couples, both with and without children, from the 2019 wave of the Panel Study of Income Dynamics (PSID). This dataset offers detailed information on time allocation, expenditures (including external childcare services), and socio-demographic variables. Our empirical analysis then proceeds in two stages. First, we estimate the childcare technology and determine the price of productive childcare time, enabling us to calculate the time cost of children and define a composite measure of full leisure that includes the leisure-like (or labor-like) component of childcare activities. Second, we estimate full leisure demand functions to derive the full cost of raising children in a Rothbarth-like approach. Our estimates indicate that, depending on the specification, 32–66% of mothers' childcare tasks are effectively productive childcare time, compared with 20–42% for fathers, implying the price of productive childcare time is substantially below market wages. For a dual-earner couple with one child, the average time cost of raising children is estimated at approximately \$230–\$570 per week at the median of the sample while the monetary cost at approximately \$620–\$890 per

[3]Market work does not enter utility directly and therefore provides the natural benchmark. Childcare with no direct utility effect is equivalent to market work, while positive (negative) process benefits make it more leisure-like (labor-like). The work–leisure distinction is therefore continuous rather than binary.

week. However, the monetary cost estimates are more fragile, for reasons discussed below.

To the best of our knowledge, this is the first study to use endogenously determined shadow prices to estimate the time cost of children. These prices provide a theoretically consistent alternative to market wages, which need not reflect the value of parental childcare time. This contrasts with existing approaches, which either value time at market wages or focus exclusively on consumption expenditures, thereby capturing only the monetary component of the cost of children. Our procedure then recovers the full cost of children (the monetary cost is obtained residually as the difference between the full cost and the time cost). While somewhat similar to Rothbarth-like approaches — which compare adult consumption between couples with and without children — our method to recover the full cost of children introduces several key innovations. The 'adult good' we analyze is parents' full leisure, which we define to include a portion of childcare time, with this fraction determined by a childcare technology function. Moreover, the demand equation is conditioned on full expenditure, i.e., total expenditure inclusive of the opportunity cost of leisure, rather than on conventional total expenditure. Our approach thus differs from approaches which focus on consumption goods rather than time use.[4] Moreover, unlike Bargain and Donni (2012) and Bargain, Donni, and Hentati (2022), we do not rely on data from single individuals and, unlike Dunbar, Lewbel, and Pendakur (2013), we refrain from imposing strong parametric restrictions on inter-individual preferences, while retaining their Similarity Across Types (SAT) condition. Conversely, our contribution can be related to the few studies that have attempted to estimate process benefits, though not specifically in the context of childcare. Graham and Green (1984) were pioneers in this area, extending Gronau's (1977) classical domestic production model while maintaining perfect substitutability between market goods and household production, but they did not address identification issues. Kerkhofs and Kooreman (2003), using a closely related framework, provide a thorough analysis of the identification of process benefits and show that identification can be achieved under certain configurations; Gørtz (2011) further extends this analysis. These studies, however, do not consider the additional complexities introduced by the presence of children. Cosaert and Hennebel (2023) are closer to our analysis because

[4]Our approach also differs from that of Arduini (2024), who relies on measures of leisure time but does not focus on children's cost.

they explicitly consider process benefits in the context of childcare. They show that process benefits are set-identified rather than point-identified, but their revealed-preference approach is fundamentally different from ours. Moreover, their objective is not to estimate the full cost of children or either of its time and monetary components.[5]

The rest of this paper is as follows. Section 2 presents the theoretical model. Section 3 outlines the empirical specification and the data. Section 4 describes the estimation method and the empirical results. Finally, section 5 concludes.

# 2 The Model

## 2.1 Decision-making in a couple with children

We consider a two-adult household, consisting of a wife ($W$) and a husband ($H$), with $n$ children, who make decisions about leisure, childcare, and consumption in a static framework.[6] The wife's and husband's leisure time and consumption are respectively denoted by $l_W$, $l_H$, $c_W$ and $c_H$ while their childcare time are respectively denoted by $t_W$ and $t_H$ and the children's consumption by $c_K$. Spouses have specific preferences for how they allocate their time and consumption. More precisely, each spouse $I$, with $I = W$, $H$, has a utility function of the form:

$$U_I = U_I(l_I, t_I, c_I, c_K, n),$$

where $U_I$ is a differentiable function, strictly increasing in $l_I$, $c_I$, and $c_K$, and strongly concave in $l_I$, $t_I$, $c_I$ and $c_K$. This specification generalizes the utility function used by Blundell, Chiappori and Meghir (2005). It allows for general interactions between parental childcare time and consumption. In particular, the utility is increasing in children's consumption, suggesting that

[5]Hallberg and Klevmarken (2003) use direct survey questions to elicit the process benefits associated with time spent on childcare.

[6]The terms 'wife' and 'husband' are used here for simplicity, and the partners are not necessarily married. As for the variable $n$, it may more generally be interpreted as a vector that also includes child characteristics such as age and sex.

parents are altruistic toward their children. Parental childcare time may affect utility through two distinct channels.[7] First, parents may derive direct utility or disutility from performing the childcare activity itself. Second, because they are altruistic toward their children, they may value the consequences of their childcare time for their children. In what follows, we use the term process benefits in a broader sense than usual, encompassing not only the direct utility from performing childcare but also the utility parents derive from its consequences for their children.[8] When the process benefits associated with childcare are positive, $\partial U_I/\partial t_I > 0$, and childcare is leisure-like at the margin. When they are non-positive, $\partial U_I/\partial t_I \leq 0$, and childcare is labor-like at the margin.[9] Finally, we assume that $\lim_{c_I \to 0} \partial U_I/\partial c_I = \lim_{c_K \to 0} \partial U_I/\partial c_K = +\infty$, which ensures that consumption is always strictly positive in equilibrium; and $\lim_{l_I \to 0} \partial U_I/\partial l_I = +\infty$ with $\partial U_I/\partial t_I < \partial U_I/\partial l_I$. This latter restriction ensures that the implicit price of parental childcare time is positive, as required by the childcare cost-minimization problem introduced below. Together with $\lim_{l_I \to 0} \partial U_I/\partial l_I = +\infty$, this condition ensures that leisure time is always strictly positive in equilibrium.

For a given number of children $n$, we assume that parents must meet some given childcare requirement. This requirement can be satisfied through parental childcare time, externally purchased childcare services, or a combination of the two. In the linear case, parental and market-provided childcare are perfect substitutes in meeting this requirement: an additional hour of parental childcare then displaces a fixed amount of purchased childcare. Parental childcare has then a clear productive dimension in the sense of Reid's (1934) third-party criterion, since it provides

[7] This feature generalizes the model proposed by Donni and Vil (2024). The few labor supply models that incorporate domestic production generally assume perfect substitutability between non-market working time and market working time (Gronau, 1977; Donni, 2008). This assumption is particularly strong and often unrealistic when applied specifically to childcare time rather than general non-market working time.

[8] For simplicity, the baseline model abstracts from domestic production beyond childcare, public consumption, externalities, mutual altruism, and cross-effects of parental childcare time on spouses' utility. Such cross-effects, in particular, may naturally arise from parental altruism toward children. We show, however, that the tractable specification used below remains compatible with these features under additional assumptions.

[9] More precisely, childcare time is locally equivalent to market work when $\partial U_I/\partial t_I = 0$. When $\partial U_I/\partial t_I < 0$, childcare is more unpleasant than market work, since it reduces utility directly in addition to displacing leisure.

a service that can alternatively be supplied by a third party, which is distinct from the utility parents may derive from performing childcare itself.[10] In the general case, the replacement rate may vary with the allocation of parental childcare time, in particular, because the amount of purchased childcare displaced by an additional hour of parental care may depend on how the parents can coordinate with each other. For a given number of children, let $g(t_W, t_H, n)$ denote the minimum expenditure on external childcare services required for given amounts of parental childcare time. The childcare requirement is represented by

$$m - g\left(t_W, t_H, n\right) \geq 0, \tag{1}$$

where $m$ is the money dedicated to purchase external childcare services. We assume that $g$ is differentiable, strictly decreasing and strictly convex in its first two arguments — which guarantees that isoquants between $m$ and $t_W$ or $t_H$ are decreasing and convex with respect to the origin — and increasing in its last argument. Additionally, we assume that

$$m \geq 0, \tag{2}$$

which ensures that external childcare expenditures can be zero or positive but never negative. For simplicity, we rule out the possibility of zero childcare time, by assuming $\lim_{t_W \to 0} \partial g/\partial t_W = \lim_{t_H \to 0} \partial g/\partial t_H = -\infty$.

A few remarks are in order. First, rather than introducing a separate childcare output and modeling its endogenous choice by parents, we adopt a more parsimonious approach that directly characterizes substitution between parental time and purchased childcare. In a more general model with an explicit childcare output entering parental utility functions, this substitution would generally depend on the endogenously chosen level of the childcare output and therefore indirectly on household resources and input prices. Our conditional representation is thus more

[10] As previously said, parental childcare may also improve child well-being in ways valued by parents. These benefits enter parental utility and thus are not included in the productive cost. The latter is defined by the substitution with external childcare only.

restrictive, but also simpler. Second, the baseline formulation does not distinguish childcare provided jointly by both parents from childcare provided by either parent alone. It therefore abstracts from any efficiency gains or losses associated specifically with joint care. This restriction is not essential, however, as joint childcare can be incorporated into the technology under fairly mild conditions.[11] Third, childcare time, $t_W$ and $t_H$, is defined strictly as the time actively dedicated to caring for children, that is, time that is explicitly taken away from either paid work or leisure. This definition excludes passive supervision, such as merely being present while performing other activities, because such time is assumed to entail a negligible opportunity cost for parents.[12] Under this assumption, the household traditional budget constraint — with the prices of both parents' and children's consumption goods normalized to one — is of the form:

$$Y - c_W - c_H - c_K - l_W w_W - l_H w_H - t_W w_W - t_H w_H - m \geq 0 \tag{3}$$

where $Y = T \cdot (w_W + w_H) + y$ is the household full income, $T$ the total time endowment of each spouse, $y$ other nonlabor income, and $w_W$ and $w_H$ the spouses' wage rates.

The decision process is assumed to lead to Pareto efficient outcomes. For a couple with children,

[11] If joint childcare can be expressed as a function of the parents' total childcare times, the function $g$ can be interpreted as the reduced form of a more general technology $g^*$ distinguishing joint from individual childcare:

$$g(t_H, t_{W,} n) = g^*(t_H - J(t_H, t_W, m, n), t_W - J(t_H, t_W, m, n), J(t_H, t_W, m, n), n)$$

where $j = J(t_H, t_{W,} m, n)$ denote the amount of childcare jointly provided by both parents, with $0 \leq J(t_H, t_W, m, n) \leq \min(t_H, t_W)$. This assumption is unlikely to be restrictive insofar as parents' total childcare times summarize the effects of the exogenous variables determining the allocation of childcare.

[12] As recently emphasized by Cuevas-Ruiz et alii (2025), passive supervision may still impose costs by restricting parents' location, availability, or simultaneous activities, or by reducing the quality of leisure and productivity of other tasks. We abstract from these costs and assume they are small relative to those of active childcare. As discussed below, however, the PSID uses a broader measure of childcare that includes passive supervision, complicating the empirical implementation of our framework.

the optimization problem of the household is thus:

$$\max_{l_W,l_H,t_W,t_H,c_W,c_H,c_K,m} \phi_W U_W(l_W,t_W,c_W,c_K,n) + \phi_H U_H(l_H,t_H,c_H,c_K,n) \tag{$\bar{\text{P}}$}$$

with $\phi_W + \phi_H = 1$, subject to the childcare constraint (1), the non-negativity constraint (2), the budget constraint (3) and time feasibility conditions: $T - l_W - t_W \geq 0$ and $T - l_H - t_H \geq 0$. In this problem, $\phi_W$ and $\phi_H$ are Pareto weights that may generally depend on all the exogenous variables. They determine the location along the Pareto frontier. If $\phi_W = 0$ (or $\phi_H = 1$), then the household behaves as though the husband always gets his way, whereas, if $\phi_W = 1$ (or $\phi_H = 0$), it is as if the wife is the effective dictator. If Pareto weights are constant, the optimization problem simplifies to the maximization of a separable household utility function, consistent with the unitary approach. If instead Pareto weights depend on wage rates, other exogenous incomes, or distribution factors — i.e., variables that affect bargaining power without entering the budget constraint — the optimization problem corresponds to the collective approach stricto sensu. Finally, the expenditure allocated to the child reflects the interaction between parents' altruism and their relative bargaining power.

As is common with collective models (see Donni and Chiappori, 2011), the optimization problem can be decentralized. If we assume that the time feasibility conditions are not binding at the optimum (implying that market work time is strictly positive), we have the following result.

**Proposition 1.** *Assume that time feasibility conditions are not binding. Then the optimal allocation of time and consumption in Problem $\bar{P}$ can be represented as the solution of a decentralized decision process. More precisely, there exists a unique pair of positive functions $(\theta_W, \theta_H)$ defined on $(w_W, w_H, Y, n)$ such that:*

(a) *the time cost is minimized:*

$$e_T(w_W,w_H,Y,n) = \min_{t_W,t_H} \left(t_W\theta_W w_W + t_H\theta_H w_H + g\left(t_W,t_H,n\right)\right) \tag{$\text{P}_0$}$$

*subject to* $g\left(t_W,t_H,n\right) \geq 0$

(b) *the household utility is maximized:*

$$\max_{\kappa_H,\kappa_W,c_K} \{\phi_W V_W(\kappa_W, c_K; w_W, \theta_W, n) + \phi_H V_H(\kappa_H, c_K; w_H, \theta_H, n)\} \tag{P$_1$}$$

*subject to* $\kappa_W + \kappa_H + c_K \leq Y - e_T$, *where*

$$V_I(\kappa_I, c_K; w_I, \theta_I, n) = \max_{l_I,t_I,c_I} U_I(l_I, t_I, c_I, c_K, n) \tag{P$_I$}$$

*subject to* $\kappa_I - c_I - l_I w_I - t_I(1-\theta_I)w_I \geq 0$, *for* $I = H, W$.

The proofs are in Appendix A. Focusing on interior solutions for market work time is important because it ensures that wage rates are observed and represent a convenient measure of the price of time.[13] Intuitively, the spouses first minimize the time cost required to satisfy the childcare constraint, with $\theta_I w_I$ as the price of the productive component of parental childcare time (Problem P$_0$). They then allocate their remaining resources between themselves and their children's consumption according to the functions $\kappa_W$, $\kappa_H$ and $c_K$ in order to maximize the household utility function (Problem P$_1$), using indirect utility functions obtained from individual optimization problems (Problems P$_H$ and P$_W$). In the individual problems, childcare time is valued at $(1-\theta_I)w_I$. Each unit of childcare time by spouse $I$ is therefore decomposed into a productive childcare component valued at $\theta_I w_I$ and a component that directly affects parental utility, valued at $(1-\theta_I)w_I$. The latter is a leisure-like activity when $\theta_I < 1$ and a labor-like activity when $\theta_I \geq 1$. If $\theta_I = 1$, there is no process benefits; similar to market labor time, childcare time simply reduces parents' leisure time.[14] Taken together, the prices $w_I^* = \theta_I w_I$ and $\bar{w}_I^* = (1-\theta_I)w_I$ can be interpretated as Lindahl prices, similar to those encountered in public

[13] As explained in the proof, however, this result extends to the case in which the time feasibility conditions are binding. The observed wage rates are then replaced by shadow wage rates, and full income is adjusted accordingly. Although the decentralized representation remains unchanged, the shadow wages are not directly observed and must be recovered, which complicates identification of the prices of time.

[14] The case where $\theta_I \leqslant 0$ is excluded by the assumption $(\partial u_I/\partial t_I)/(\partial u_I/\partial l_I) < 1$. Without this assumption, the analysis would be significantly more complicated.

economics,[15] and satisfy $w_I^* + \bar{w}_I^* = w_I$.

The time cost of children, defined as $e_T(w_W, w_H, Y, n)$, is evaluated using the prices $w_W^* = \theta_W w_W$ and $w_H^* = \theta_H w_H$, rather than the standard wage rates. If childcare is leisure-like, so that $w_I^* < w_I$, valuing parental time at market wages overstates the time cost of children, ceteris paribus. Conversely, if childcare is labor-like, so that $w_I^* \geq w_I$, market-wage valuation is exact when $w_I^* = w_I$ and understates the time cost when $w_I^* > w_I$. The full cost is defined as the sum of the time cost $e_T(w_W, w_H, Y, n)$ and the monetary cost $e_M(w_W, w_H, Y, n)$, the latter being equal to $c_K(w_W, w_H, Y, n)$.

Finally, it is important to note that this result continues to hold when the constraint (2) is binding, i.e., when the household does not purchase external childcare services. In that case, the Kuhn–Tucker multiplier associated with this constraint may be positive, giving rise to an adjustment factor $\nu \geq 1$, with $\nu = 1$ whenever $m > 0$.[16] This factor affects the marginal conditions characterizing the optimal mix of parental childcare times when $m = 0$ since the prices of productive childcare time can then be written as $w_I^* = \nu\theta_I w_I$. Importantly, however, the factor $\nu$ does not alter the decomposition of the price of parental time into $\theta_I w_I$ and $(1-\theta_I)w_I$, nor does it enter the time cost of children. Indeed, the price-scale coefficient $\theta_I$ is determined by the trade-off between childcare time and leisure in parental preferences and measures the fraction of the opportunity cost $w_I$ attributed to the productive component of childcare. The factor $\nu$, by contrast, arises solely because the household is prevented from choosing negative expenditures on external childcare when $m = 0$.

## 2.2 Identification and Empirical Tractability

The primary objective of this study is to show how the full cost of children — and its decomposition into time cost and monetary cost — can be identified from traditional survey data. The childcare technology, and hence the time cost of children, can be identified under fairly general

[15] See also Donni (2007, 2009) for an application to the household.

[16] The details are given in the proof of Proposition 1 in Appendix.

conditions. Identifying the remaining components of the model requires additional restrictions that will be detailled below.

### 2.2.1 Identification of the Time Cost of Children

As is usual in the literature on the identification of collective models, we first assume that individual consumptions $c_W$, $c_H$ and $c_K$ are not observed by the economist. Instead, only their sum $c_W+c_H+c_K$ is observable. The other dependent variables, $l_W, t_W, l_H, t_H$ and $m$, are observed as functions of the exogenous variables. The optimal solutions of the cost minimization problem are denoted by $m^*$, $t_W^*$ and $t_H^*$ each expressed as a function of $(w_W, w_H, Y, n)$.

The first identification result is the following.

**Proposition 2.** *Assume that time feasibility conditions are not binding. Under regularity and support conditions, the childcare technology can be recovered from the observation of* $m^*$, $t_W^*$, *and* $t_H^*$ *as functions of* $(w_W, w_H, Y, n)$. *The scalar* $\theta_W$ *and* $\theta_H$ *are exactly identified when* $m^* > 0$, *and are identified up to a common multiplicative function when* $m^* = 0$.

Intuitively, when $m^* > 0$, the prices of productive childcare time, $w_H^* = \theta_H w_H$ and $w_W^* = \theta_W w_W$, correspond to the marginal rate of technical substitution between $t_W$ and $t_H$ on the one hand, and $m$ on the other. From these marginal rates of substitution, the corresponding price-scale coefficients $\theta_H$ and $\theta_W$ are then directly recovered. When $m^* = 0$, the non-negativity constraint on $m$ is binding, so that these marginal rates identify $\nu w_H^*$ and $\nu w_W^*$, with a common factor $\nu \geq 1$. Hence, only the ratio of the price-scale coefficients is identified nonparametrically. Full identification requires further restrictions that will be discussed below.

Identifying the other component of the full cost of children raises additional difficulties. The general principle underlying almost all existing approaches is to compare the behavior of couples with children with that of otherwise similar couples without children. More formally, these approaches rely on the existence of one or more stable relationships among at least two observable variables that are invariant to the presence of children once the resources devoted to them have been accounted for, and that can be recovered from observed behavior. The full cost of children is

then identified as the resource adjustment required to restore these relationships when comparing otherwise similar couples with and without children. To derive these stable relationships, we proceed in two stages, developed in the next subsections.

### 2.2.2 The Full Leisure Demand Functions

The decision process described in Proposition 1 is not strictly speaking sequential as the first-stage Problem $P_0$ depends on the prices of productive childcare time, which are also determined by the second-stage Problems $P_1$. To obtain a clear sequential decision framework, we adopt a classical assumption used in all models that incorporate process benefits (Gørtz, 2011; Green and Graham, 1984; Kerkhofs and Kooreman, 2003; Cosaert and Hennebel, 2023; Donni and Matteazzi, 2018), imposing additional structure on the utility functions as described below.

**Assumption A.1** Each spouse $I$ has a utility function of the form:

$$U_I = U_I(l_I + \varphi_I(t_I), c_I, c_K, n)$$

with $I = W$ or $H$, where $L_I = l_I + \varphi_I(t_I)$ can be viewed as the 'full leisure' of spouse $I$ for some differentiable and concave functions $\varphi_I(\cdot)$, satisfying $\varphi_I(0) = 0$.

This additional structure preserves the core properties of our model while making it more tractable for empirical estimation. It offers several advantages. Firstly, leisure time and leisure-like (or labor-like) childcare time are combined into a single composite full leisure time. This definition can be applied consistently to both couples with and without children, facilitating comparisons between them. Secondly, the decision process is a sequential one. Specifically, the price of the leisure-like (or labor-like) activity in the second-stage Problem $P_1$ is determined by the first-stage choice of childcare time only, as detailed in the following proposition. Thirdly, the assumption provides the additional structure needed to fully identify the price-scale coefficients, thereby resolving the multiplicative indeterminacy that would otherwise arise when external childcare expenditure is zero.

The following proposition first establishes the sequential structure of the decision process.

**Proposition 3.** *Assume that time feasibility conditions are not binding. Then, under Assumption A.1, the optimal allocation of time and consumption in Problem $\bar{P}$ is sequential. Firstly, childcare time is determined by the Problem $P_0$ with $\theta_W = 1 - \varphi'_W(t_W)$ and $\theta_H = 1 - \varphi'_H(t_H)$, giving $t^*_H(w_W, w_H, n)$ and $t^*_W(w_W, w_H, n)$ as optimal levels of childcare time. In particular, both optimal childcare times and the associated implicit price parameters are independent of nonlabor income $Y$. Secondly, the remaining resources are allocated by the Problem $P_1$, where each spouse's indirect utility is generated by the corresponding Problem $P_I$, giving $\kappa_H$, $\kappa_W$, and $c_K$. In Problem $P_I$, spouse $I$ maximizes her or his own utility function $U_I(L_I, c_I, c_K, n)$ with respect to $(L_I, c_I)$ subject to the budget constraint $\kappa_I + \pi_I - c_I - L_I w_I \geq 0$, giving $L^*_I = L_I(w_I, \kappa_I + \pi_I, c_K, n)$ as optimal levels of full leisure, with $\pi_I = \varphi_I(t^*_I)w_I - t^*_I(1 - \theta_I)w_I$ and $\theta_I = 1 - \varphi'_I(t^*_I)$.*

This result has several useful implications. First, the prices of parental childcare time are determined entirely at the cost-minimization stage and, apart from the functions $\varphi_I$, independently of the remaining features of individual preferences.[17] Second, the resulting childcare demand functions are independent of nonlabor income $Y$ and, more generally, of variables that affect utility without entering the childcare constraint. This constitutes a strong and testable implication of the model. Once childcare time has been determined, spouse $I$'s utility $U_I(L_I, c_I, c_K, n)$ is maximized subject to the budget constraint: $c_I + L_I w_I \leq \kappa_I + \pi_I$. In this representation, the function $\pi_I$ can be viewed as a profit function associated with the production of full leisure: full leisure is valued at the wage rate $w_I$, while childcare time enters as an input with implicit price $\overline{w}^*_I = (1 - \theta_I)w_I$. In the empirical application, we will adopt an even more simplified formulation in which the functions $\varphi_I(t_I)$ are linear in $t_I$, so that the functions $\pi_I$ reduce to zero. This specification also admits a particularly intuitive interpretation: childcare time can be decomposed into a productive domestic-work component and a leisure-equivalent component associated with process benefits. We return to this interpretation in the empirical section.

The additional structure also has useful implications for identification.

[17] This concept is known as the separation principle in agricultural economics (Benjamin, 1992).

**Corollary 4.** *Under Assumption A.1 and the conditions of Proposition 2, the functions* $\theta_I(t_I) = 1 - \varphi_I'(t_I)$*, for* $I = W, H$*, are identified over the entire relevant support of parental childcare time. Consequently, when* $m^* = 0$*, the multiplier* $\nu$ *is identified from*

$$\nu = -\frac{\partial g/\partial t_I}{\theta_I(t_I^*)w_I}, \qquad I = W, H.$$

The idea is that variation in the region where $m^* > 0$ identifies $\theta_I(t_I)$ at all childcare time values that are relevant when $m^* = 0$. Once the functions $\theta_I(t_I)$ are identified over the relevant support, the common multiplier $\nu$ can be recovered directly from the first-order conditions. The fact that either spouse yields the same value of $\nu$ also provides a restriction of the model.

### 2.2.3 Identification of the Monetary Cost of Children: General Case

By application of the previous results, the technology $g$ and the price-scale coefficients $\theta_I$ are identified. Since $\theta_I = 1 - \varphi_I'(t_I)$, the identified price parameter can be expressed as a function of $t_I$ alone. The function $\varphi_I(t_I)$ can then be recovered by integrating $1 - \theta_I(t_I)$, using the boundary condition $\varphi_I(0) = 0$. Once $\varphi_I$ is recovered, full leisure time functions can then be constructed as $L_I = l_I + \varphi_I(t_I)$ and the corresponding profit function $\pi_I$ can be recovered as well. From now, we assume that optimal solutions $L_W^*$ and $L_H^*$ as well as the utility profit and the time cost are observed as functions of $(w_W, w_H, Y, n)$.

The remaining structural components of the model, and in particular children's consumption $c_K$ can also be identified. The basic idea is to identify the full cost of children and recover the monetary cost as the residual after the time cost has been identified. This strategy, however, requires additional restrictions. We therefore impose a further assumption stating that some dimensions of parental preferences are invariant to the presence of children.

**Assumption A.2.** Individual utility functions are separable between adult-related variables and child-related variables, so that

$$U_I = U_I(u_I(L_I, c_I), c_K, n),$$

with $I = W$ or $H$, where $u_I$ is a differentiable and concave function.

This type of assumption, in one form or another, is standard in the literature. Blundell, Chiappori, and Meghir (2005) impose a very similar separability condition. Gronau (1991), Bargain and Donni (2012), and Bargain, Donni, and Hentati (2022) derive a related separable structure from caring preferences, whereby parents care about their children's utility rather than directly about the goods they consume, thereby separating parents' own consumption and leisure from child-related arguments. Technically, it simply implies that the marginal rate of substitution between $L_I$ and $c_I$ is independent of both the number of children and their consumption. This invariance is crucial for identification, as it provides the stable relationship between the behavior of couples with and without children that is needed to recover children's consumption.

However, this assumption may be regarded as excessively strong and it can be be relaxed and replaced, for instance, by the following alternative.

**Assumption A.2′** Individual utility functions take the form:

$$U_I = U_I(u_I(L_I, c_I, n), c_K, n),$$

with $I = W$ or $H$, where $u_I$ is a differentiable and concave function, such that children affect the corresponding demand for full leisure only through an additive demographic term:

$$L_I = \alpha_I(w_I, n) + \beta_I(w_I, \kappa_I).$$

for some functions $\alpha_I(w_I, n)$ and $\beta_I(w_I, \kappa_I)$.

This specification implies that the number of children leaves the slope of the Engel curve unchanged while shifting its intercept — an assumption often referred to as Similarity Across Types (SAT). Although still restrictive, it is weaker than Assumption A.2, which requires the whole relationship to be identical for households with and without children. The child-specific term $\alpha_I(w_I, n)$ may capture differences in preferences between parents and non-parents, as well as economies of scale associated with raising children.

In what follows, we will use these two assumptions. The remaining difficulty is that individual consumptions $c_H$ and $c_W$ are not separately observed, preventing the direct exploitation of a stable relationship between these variables and full leisure $L_H$ and $L_W$. From standard results in the literature (Chiappori, 1988, 1992, or Chiappori, Fortin and Lacroix, 2002, for instance), however, individual consumptions as well as the sharing functions $\kappa_W$ and $\kappa_H$ can be identified up to a unique constant from a sample of childless couples provided that certain regularity conditions are satisfied. Under the stability condition in Assumption A.2, the full-leisure demand functions identified from childless couples also apply to couples with children. Hence,

$$L_I = L_I(w_I, \kappa_I(w_H, w_W, Y_R, n) + k_I),$$

for couples with children, where the functions $L_I(\cdot)$ are identified, while $k_W$ and $k_H$ represent the remaining indeterminacy in the sharing rule and satisfy $k_W + k_H = 0$. The spouses' resource shares satisfy the adding-up condition:

$$Y + \pi_W + \pi_H - e_M - e_T = \kappa_W + \kappa_H.$$

Provided that each full leisure demand function can be inverted with respect to its second argument, we obtain:

$$\kappa_I + k_I = L_I^{-1}(w_I, L_I).$$

Summing these expressions across spouses eliminates the unidentified constants because $k_W + k_H = 0$, yielding:

$$Y + \pi_W + \pi_H - e_M - e_T = L_W^{-1}(w_W, L_W) + L_H^{-1}(w_H, L_H)$$

Hence, although the individual sharing functions are identified only up to an additive constant, their sum — and therefore the resources remaining after accounting for children — is exactly identified. Since $Y$, $\pi_W$, $\pi_H$, and the time cost $e_T$ are known, this identity identifies the monetary

cost $e_M$. Consequently, all components of the full cost of children are identified under the stated regularity and stability conditions.

The identification result above relies on the strong preference restriction embodied in Assumption A.2. An alternative is to relax this assumption and instead exploit nonlinearities in the full leisure demand functions. Specifically, if Assumption A.2 is replaced by Assumption A.2′, the inversion of the demand function is replaced by the inversion of its derivative with respect to remaining full income. Formally, let

$$D_I(w_I, \kappa_I + k_I) = \frac{\partial L_I}{\partial(\kappa_I + k_I)}$$

denote the marginal response of full leisure to individual resources. This function is independent of the number of children $n$. If $D_I(w_I, \cdot)$ is invertible, then

$$\kappa_I + k_I = D_I^{-1}\left(w_I, \frac{\partial L_I}{\partial(\kappa_I + k_I)}\right).$$

The remainder of the proof then proceeds exactly as above. We then present the formal result.

**Proposition 5.** *Under Assumption A.1, the conditions of Proposition 2 and the regularity conditions ensuring identification of the collective labor supply model (Chiappori, 1988, 1992), the full cost of children is identified from the observation of the full leisure demand functions $L_W^*$ and $L_H^*$ as functions of $(w_W, w_H, Y, n)$, provided that either: (i) Assumption A.2 holds and $L_W$ and $L_H$ are invertible with respect to individual shares; or (ii) Assumption A.2′ holds and the corresponding marginal response functions $D_W$ and $D_H$ are invertible.*

The identifying assumptions generate several testable implications. First, the collective labor-supply structure itself imposes the usual restrictions derived by Chiappori (1988, 1992), which can be tested independently of the additional stability assumptions introduced here. Second, the additional stability restrictions are themselves testable through nested specifications. Assumption A.2 implies that full-leisure demand is invariant to $n$, whereas A.2′ allows level shifts but requires marginal responses to individual resources to remain invariant. Thus, A.2 can be

tested against A.2′, and A.2′ against a more general alternative.

### 2.2.4 A More Tractable and Robust Approach

The previous approaches are quite general, but they rely on the regularity conditions required for the identification of a collective labor-supply model. In particular, identification requires sufficiently rich nonlinearities in both behavioral relationships.[18] Since our primary objective is to identify the cost of children rather than the exact allocation of full income between spouses, we adopt a more tractable strategy based on the estimation of reduced-form equations and impose the following assumption.

**Assumption A.3** The sharing functions are invariant to the presence of children:

$$\kappa_I(w_H, w_W, Y_R) = \kappa_I(w_H, w_W, Y_R, n),$$

with $I = W, H$, where $Y_R$ denotes remaining full income, defined by $Y_R = Y$ for a couple without children and $Y_R = Y + \pi_W + \pi_H - e_M - e_T$ for a couple with children.

This assumption is automatically satisfied under the unitary approach, in which bargaining weights are constant, but it is more general than that special case. Taken together, Assumptions A.1-A.3 imply that the presence of children affects full leisure only through a pure income effect, without altering individual preferences or the intra-household allocation rule between parents. That is, children may modify each parent's share of full income only insofar as it changes $Y_R$ itself — that is, in exactly the same way as a change in full income would. This is the central intuition underlying the traditional Rothbarth approach. In addition, unlike the preceding identification result, which relies closely on the collective framework of Chiappori (1988, 1992), the reduced-form representation derived below under Assumption A.3 is robust to a broad range of interactions between spouses' choices that are not explicitly modeled here, including those arising from mutual altruism, public consumption, consumption externalities, or

[18] As an alternative, identification may rely on the existence of valid distribution factors, which is also a restrictive requirement (Chiappori, Fortin and Lacroix, 2002).

domestic production. For the purpose of identifying the full cost of children, these interactions need not be separately modeled or structurally identified, provided that they do not cause the presence of children to affect behavior independently of remaining full income.

Then we simply assume that the reduced-form full leisure demand functions $F_I(w_H, w_W, Y_R)$, defined as

$$L_I = L_I(w_I, \kappa_I(w_H, w_W, Y_R)) = F_I(w_H, w_W, Y_R),$$

can be identified from a sample of childless couples.[19] These reduced-form functions are invariant to the presence of children; they therefore apply to both childless couples and couples with children and can consequently be identified from a sample of childless couples. For couples with children, remaining full income $Y_R$ is equal to $Y + \pi_W + \pi_H - e_M - e_T$. Hence,

$$L_I = F_I(w_H, w_W, Y + \pi_W + \pi_H - e_M - e_T), \tag{4}$$

where $L_I = l_I + \varphi_I(t_I)$. If $F_I(w_H, w_W, \cdot)$ is invertible, the reduced-form demand function can be inverted with respect to the remaining full income, which allows for exact identification of the full cost of children. Specifically, we obtain:

$$e_M = Y + \pi_W + \pi_H - e_T - F_I^{-1}(w_H, w_W, L_I), \tag{5}$$

where all the terms on the right-hand side are either observable or identifiable. Apart from the specific definition of full leisure adopted here, this approach is closely related to that proposed by Bradbury (2008). Identification requires the regularity condition that the Engel curve for full leisure is locally invertible with respect to remaining full income. Unlike the standard identification of collective labor supply models, however, no additional nonlinearity condition is required, which makes the approach considerably simpler to implement. In addition, the full

[19] Under the baseline collective structure, the functions $F_I(\cdot)$ inherit the restrictions derived by Chiappori (1988, 1992). These restrictions need not hold under the more general interactions discussed above and are therefore not imposed or discussed in what follows.

cost of children can be identified from a single spouse's full leisure demand function.

The same result can be obtained under Assumption A.2′. The reduced-form full leisure demand function is:

$$F_I(w_H, w_W, Y_R, n) = \alpha_I(w_I, n) + \gamma_I(w_H, w_W, Y_R),$$

where $\gamma_I(w_H, w_W, Y_R) = \beta_I(w_I, \kappa_I(w_H, w_W, Y_R))$. Under this specification, the level of the demand function no longer identifies remaining full income, and hence the full cost of children, because of the unknown demographic term $\alpha_I(w_I, n)$. Identification instead relies on the marginal response of full leisure to remaining full income. Define:

$$D_I(w_H, w_W, Y_R) = \frac{\partial F_I(w_H, w_W, Y_R, n)}{\partial Y_R} = \frac{\partial \gamma_I(w_H, w_W, Y_R)}{\partial Y_R}.$$

Since the additive term $\alpha_I(w_I, n)$ disappears upon differentiation, $D_I$ is invariant to the presence of children. If $D_I(w_H, w_W, \cdot)$ is locally invertible, then identification proceeds as before. This alternative strategy avoids the estimation of a complete collective labor supply model and requires only a single reduced-form full leisure demand equation. Its drawback is that identification rests entirely on the curvature of this equation. When curvature is weak or imprecisely estimated, the resulting identification may therefore be fragile.

To summarize, we have the following result.

**Proposition 6.** *Under Assumptions A.1 and A.3, and the conditions of Proposition 2, the full cost of children is identified from the observation of a single reduced-form full-leisure demand function $F_I$ as function of $(w_W, w_H, Y)$, provided that either: (i) Assumption A.2 holds and the function $F_I(w_W, w_H, \cdot)$ is locally invertible with respect to remaining full income; or (ii) Assumption A.2′ holds and the corresponding marginal-response function $D_I(w_W, w_H, \cdot)$ is locally invertible.*

As discussed above, this reduced-form strategy is robust to a broad range of interactions between spouses' choices that need not be explicitly modeled or structurally identified. A further advan-

tage is that identification requires only one spouse's full-leisure demand function. Estimating both spouses' demand functions may nevertheless improve precision and yields overidentifying restrictions that can be used for specification testing.[20]

# 3 Data and Empirical Specification

In this section, we first provide an overview of the data and then specify functional forms for the childcare technology and individual preferences.

## 3.1 Data

This subsection outlines the sample selection process and describes the variables used in the estimation. We also provide some preliminary, intuitive results based on reduced-form equations.

### 3.1.1 Sample selection

Our empirical analysis relies on the 2019 wave of the Panel Study of Income Dynamics (PSID), a longitudinal survey that provides detailed information on a broad range of topics, including income and wealth sources, wage rates, working hours, and socio-demographic characteristics. This wave also includes data on household members' time allocation — covering leisure, personal care, and childcare[21] — as well as disaggregated household expenditures, notably parents' spending on external childcare services. The 2019 wave covers 9,569 households. For our analysis, we focus on non-student heterosexual couples aged 22 to 50, with and without children and exclude households with more than two children and those in which the youngest child is older than six years. Because our empirical strategy relies on observed wages, we further restrict the sample to

[20] In the empirical specification, we allow the additive demographic term to depend on both spouses' wage rates. This is more general than the decentralized representation above, without altering the identification results, which rely only on its additive separability from remaining full income.

[21] In the PSID, time use information is collected through retrospective questions, rather than time diaries. Nevertheless, Insolera, Johnson, and Simmert (2019) find that time use measures from the PSID are generally consistent with those reported in the American Time Use Survey (ATUS).

dual-earner couples. This latter restriction, in particular, reduces the sample by about 17%, or 20% when considering only couples with children.[22] Our final sample comprises 825 households: 463 with children and 362 without.

### 3.1.2 Key variables

The period of analysis is one week, and all variables are calibrated accordingly. The estimation of the model relies on two key time-related variables: parental childcare time and leisure time. The PSID measures parental childcare time using the following question: "In a typical week, how many hours [do you/does [he/she]] spend caring for or looking after children?" This definition is intentionally broad, encompassing time spent caring for children even when it occurs alongside other activities — including sleep. Consequently, reported childcare time can reach up to 168 hours per week, as documented in Table 1. These figures do not allow for an accurate assessment of the actual time cost of children, since a substantial share of this reported time may coincide with low-effort or non-exclusive tasks and therefore impose little to no opportunity cost on parents.[23] To address this issue, we adjust the reported values downward to match more closely the American Time Use Survey (ATUS) estimates, which capture primary childcare time, defined as time spent when childcare is the respondent's main activity. If this correction takes the form of subtracting a constant from all observations, it should not greatly affect the estimation of marginal prices, which depend on variation at the margin rather than on absolute levels. How-

[22]Restricting the sample to dual-earner couples may induce selection if non-working parents differ systematically in childcare burdens or preferences for parental care. However, we focus on ages at which labor-force participation is high, so the resulting selection is relatively limited. Our estimates should nevertheless be interpreted as applying primarily to dual-earner couples.

[23]The original PSID measures reported in Table 1 are consistent with ATUS measures when primary and secondary childcare activities are combined. According to published ATUS tables for 2015–2019 (U.S. Bureau of Labor Statistics, 2019, Tables 9 and 10), total primary and secondary childcare amounts to about 39.2 hours per week for men and 63.3 hours for women living in households with at least one child under age 6. This is consistent with Stewart and Allard (2015), who show that, among families with children, secondary childcare accounts for a large share of total childcare time, with both men and women spending nearly three times more time on secondary than on primary childcare.

ever, it might generate negative values for parents who report low overall childcare involvement in the PSID. To avoid this eventuality, we apply the following non-linear transformation:

$$t_I^{adj} = t_I^{psid} - k_I \times \tanh\left(\frac{t_I^{psid}}{k_I}\right)$$

where $t_I^{psid}$ is number of hours of child care time as reported by the PSID. Before applying the transformation, we bound this measure so that the sum of weekly childcare and market work does not exceed the 168 hours available in a week. The constant $k_I$ are chosen so that the average adjusted childcare time matches the average values reported in the ATUS.[24] This transformation has two desirable properties:[25] it preserves the zero point — i.e., $t_I^{adj} = t_I^{psid}$ when $t_I^{psid} = 0$ — and it approximates a constant downward shift for large values of reported time, since $t_I^{adj} \approx t_I^{psid} - k_I$ when $t_I^{psid}$ is large. As shown in Table 1, the proportion of zero values is negligible and remains unchanged under both definitions, while average childcare time falls substantially under the adjusted measure — to about 10 hours per week for fathers and 19 hours for mothers. Because the adjustment is non-linear, it necessarily changes the distribution of childcare time and may also alter its joint distribution across spouses. We shall therefore assess the sensitivity of our results to this choice by also estimating the model using the original, unadjusted PSID childcare measure; the corresponding results are reported in Section 4.2.1.

The second key time-related variable is leisure time. The PSID includes a direct question on usual weekly leisure hours: "In a typical week, how many hours [do you/does [he/she]] spend doing leisure activities for enjoyment, for example, watching TV, doing physical activities that [you enjoy / he or she enjoys], going online, or spending time with friends?" However, this direct measure differs from the notion of leisure used in our model, which is closer to the broader concept of non-market time. The PSID leisure measure excludes many non-market activities — such as

[24]To do so, we use time use data for men and women with child under 6 from the published ATUS tables for 2015–2019 (U.S. Bureau of Labor Statistics, 2019, Tables 9). In these aggregated data, childcare time refers exclusively to primary activities. It includes physical care, education-related activities, reading to or with children, playing or engaging in hobbies with children, and travel related to caring for and helping household children.

[25]Recall that $\tanh(x) = \left(e^x - e^{-x}\right) / \left(e^x + e^{-x}\right)$.

Table 1: Parents' Weekly Childcare Time (in Hours, with and without Adjustment) and Weekly Childcare Expenditure (in Dollars)

| | Mean | StDev | Min | 50% | Max | Pzeros |
|---|---|---|---|---|---|---|
| Father's Original PSID Childcare Time | 35.12 | 39.72 | 0.00 | 20.00 | 168.00 | 0.02 |
| Mother's Original PSID Childcare Time | 53.65 | 46.39 | 0.00 | 40.00 | 168.00 | 0.01 |
| Father's Adjusted Childcare Time | 9.66 | 20.65 | 0.00 | 0.91 | 115.17 | 0.02 |
| Mother's Adjusted Childcare Time | 18.97 | 28.45 | 0.00 | 6.63 | 116.91 | 0.01 |
| Childcare Expenditure | 99.91 | 127.46 | 0.00 | 49.96 | 769.23 | 0.35 |

Notes: Mean = average value, StDev = standard-deviation, Min = minimum value, 50% = median, Max = maximum value, Pzeros = proportion of zeros.

personal care, household production, shopping, and sleep — even though these uses of time may also contract when parental time dedicated to childcare increases. This matters because childcare need not crowd out narrowly defined leisure one-for-one; instead, the required time may be drawn from several components of non-market time. A measure based only on reported leisure would therefore miss part of the time reallocation induced by children. For these reasons, we follow the standard practice in the literature and define leisure time as non-market time. Specifically, weekly leisure time is computed as $l_I = T - h_I - t_I$, where $T$ is fixed at 119.7 hours[26] per week, and $h_I$ denotes hours spent in paid work. Once the $\varphi_I(t_I)$ will be estimated, it will be possible to compute the full leisure time as $L_I = l_I + \varphi_I(t_I) = T - h_I - (1 - \varphi_I(t_I))t_I$, where $\varphi_I(t_I)$ corresponds to the part of parental childcare time perceived as leisure.

Finally, the PSID elicits childcare expenditures using the question: "How much did you (and your family living there) pay for child care in 2018?" To obtain weekly childcare expenditures, we divide the reported annual amount by 52. As with other expenditure variables in the PSID, childcare expenditures refer to the previous calendar year, whereas childcare time refers to a typical week around the time of the interview, creating some temporal mismatch between the two measures. As shown in Table 1, the proportion of zeros is substantial but but does not compromise the feasibility of the estimation.

[26]This figure corresponds to $24 - 6.9 = 17.1$ hours per day, where 6.9 hours is the average sleeping time.

Table 2: Descriptive Statistics of the Benchmark Sample

| | Without Child | | With Children | |
|---|---|---|---|---|
| | Mean | StDev | Mean | StDev |
| Father's Adjusted Childcare Time | 0.00 | 0.00 | 9.66 | 20.65 |
| Mother's Adjusted Childcare Time | 0.00 | 0.00 | 18.97 | 28.45 |
| Childcare Expenditure | 0.00 | 0.00 | 99.91 | 127.46 |
| Number of Children | 0.00 | 0.00 | 1.56 | 0.50 |
| Child under 3 | 0.00 | 0.00 | 0.74 | 0.44 |
| Husband's Leisure Time | 77.03 | 12.28 | 66.46 | 22.32 |
| Wife's Leisure Time | 82.49 | 12.30 | 67.19 | 28.62 |
| Husband's Hourly Wage | 30.71 | 44.51 | 31.97 | 25.89 |
| Wife's Hourly Wage | 24.36 | 16.36 | 29.48 | 26.25 |
| Full Expenditure (in thousand dollars) | 5.52 | 4.15 | 6.31 | 4.25 |
| Husband's Education in Years | 14.24 | 2.26 | 14.38 | 2.39 |
| Wife's Education in Years | 14.84 | 2.10 | 15.15 | 1.87 |
| Husband's Age in Years | 36.71 | 7.82 | 34.21 | 5.33 |
| Wife's Age in Years | 35.58 | 7.84 | 32.88 | 5.04 |

Notes: StDev = standard-deviation. Full expenditure is defined as the sum of household total expenditure on goods and services and expenditure in leisure time computed with spouses' wage rates as prices.

Summary statistics for the variables used in the estimations are presented in Table 2, separately for couples without children and couples with children. On average, childless women report about 15 more hours of leisure per week than mothers, while childless men report around 11 more hours than fathers. For couples with children, average weekly childcare expenditures amount to approximately $100, with a large standard deviation reflecting substantial heterogeneity in spending. Aside from differences directly related to the presence of children, the two groups of couples exhibit broadly similar average characteristics across the remaining variables. These include spouses' hourly wages (directly provided by the PSID), education levels (measured as the highest grade of school completed), ages (in years). The variables also include the household full expenditure (defined as the sum of household total expenditure on goods and services and expenditure in leisure time computed with spouses' wage rates as prices) that is used in the estimation instead of the full income to ensure model consistency with a life-cycle framework

(Blundell and MaCurdy, 1999).

### 3.1.3 Preliminary regressions

Before turning to the structural model, we begin by presenting reduced-form OLS estimates of the childcare time and expenditure equations. The estimates are reported in Table 3. The effects of parental hourly wages on childcare expenditures are positive and decreasing, while the effects on childcare time are negative and increasing (that is, the curves are concave with respect to wages). These estimates allow for a back-of-the-envelope approximation of the price of parental time devoted to childcare. Evaluated at the average point of the sample (that is, $w_H = 31.97$ and $w_W = 29.48$ using figures in Table 2), a \$10 increase in the father's (mother's) hourly wage is associated with an increase of approximately \$6.44 (\$15.60) in spending on childcare services. At the same time, a \$10 increase in the father's (mother's) hourly wage is associated with a reduction of approximately 0.34 (2.05) hours in the father's (mother's) time devoted to childcare. This implies that the father substitutes one hour of childcare time with \$18.94($\approx 6.44/0.34$) of childcare services — interpreted as the value of the father's childcare time. Relative to the value of his wage, this gives $\theta_H = 0.59 \approx 18.94/31.97$. Similarly, the variation in the value of the mother's childcare time is \$7.61($\approx 15.60/2.05$), giving $\theta_W = 0.26 \approx 7.61/29.48$.

## 3.2 Empirical Specification

In this subsection, we present the functional form for the childcare technology and the demands for full leisure. We also incorporate observable and unobservable heterogeneity.

### 3.2.1 The Childcare Technology

Instead of specifying the childcare technology, we directly model the time cost function. We begin by considering the case of interior solutions.

If $m > 0$ (i.e., the solution is interior), the time cost function is as follows:

$$e_T = \alpha^* + \beta^*_W w^*_W + \beta^*_H w^*_H + 2\gamma_W \sqrt{w^*_W} + 2\gamma_H \sqrt{w^*_H} + 2\gamma_{WH} \sqrt{w^*_W w^*_H}, \tag{6}$$

Table 3: OLS estimations of reduced-form childcare time and expenditure equations

| | Expenditure | | Father's Time | | Mother's Time | |
|---|---|---|---|---|---|---|
| | Est | StErr | Est | StErr | Est | StErr |
| Constant | 31.52 | 16.72 | 12.85 | 2.94 | 26.31 | 3.77 |
| Father's Wage | 0.74 | 0.76 | -0.04 | 0.09 | -0.05 | 0.12 |
| Squared Father's Wage / 100 | -0.15 | 0.70 | 0.01 | 0.04 | 0.03 | 0.05 |
| Mother's Wage | 1.92 | 0.55 | -0.10 | 0.07 | -0.27 | 0.11 |
| Squared Mother's Wage / 100 | -0.61 | 0.33 | 0.05 | 0.03 | 0.11 | 0.04 |
| Father's Age | 3.31 | 8.41 | 0.27 | 1.80 | -0.65 | 2.27 |
| Mother's Age | 11.83 | 9.63 | -1.56 | 1.92 | -1.50 | 2.47 |
| Father's Education | 11.08 | 5.70 | -0.77 | 1.18 | -2.63 | 1.50 |
| Mother's Education | 10.18 | 5.67 | -0.15 | 1.05 | 0.49 | 1.73 |
| Number of Children | 19.47 | 5.52 | -0.57 | 1.00 | -1.19 | 1.31 |
| Dummy for Child under 3 | 6.96 | 5.11 | 0.23 | 1.04 | 0.33 | 1.39 |

Notes: All demographic variables are standardized. Est = estimate, StErr = standard-error.

where $\alpha^*$, $\beta^*_W$, $\beta^*_H$, $\gamma_W$, $\gamma_H$ and $\gamma_{WH}$ are parameters, $w^*_H$ and $w^*_W$ are the price of parents' productive childcare time as previously explained, which is supposed to be proportionate to hourly wage rates: $w^*_I = \theta_I w_I$ for some positive parameter $\theta_I$. This specification, which is inspired from the Generalized Leontieff of Diewert (1971), is globally consistent with the theory (increasing and concave with respect to prices) whenever $\beta^*_W \geq 0$, $\beta^*_H \geq 0$, $\gamma_H \geq 0$, $\gamma_W \geq 0$ and $\gamma_{WH} \geq 0$. Its appeal is that it accommodates both substitution and complementarity between parents' inputs through the interaction term $\sqrt{w^*_W w^*_H}$, albeit potentially at the cost of the global consistency.

From Shephard's Lemma, the solution of the cost minimization problem is:

$$t_H = \beta^*_H + \gamma_H \frac{1}{\sqrt{w^*_H}} + \gamma_{WH} \sqrt{\frac{w^*_W}{w^*_H}} \tag{7}$$

$$t_W = \beta^*_W + \gamma_W \frac{1}{\sqrt{w^*_W}} + \gamma_{WH} \sqrt{\frac{w^*_H}{w^*_W}} \tag{8}$$

if $m > 0$. Defining $\beta_I^* = \boldsymbol{\beta}_I \boldsymbol{z}_{T,I} + v_I$, where $\boldsymbol{z}_{T,I}$ are control variables, $v_I$ is an error term, and using $w_I^* = \theta_I w_I$, we obtain the childcare time equations:

$$t_H = \boldsymbol{\beta}_H \boldsymbol{z}_{T,H} + \gamma_H \frac{1}{\sqrt{\theta_H w_H}} + \gamma_{WH} \sqrt{\frac{\theta_W w_W}{\theta_H w_H}} + v_H \tag{9}$$

$$t_W = \boldsymbol{\beta}_W \boldsymbol{z}_{T,W} + \gamma_W \frac{1}{\sqrt{\theta_W w_W}} + \gamma_{WH} \sqrt{\frac{\theta_H w_H}{\theta_W w_W}} + v_H. \tag{10}$$

To incorporate heterogeneity in the demand for external childcare services, we similarly write: $\alpha^* = \boldsymbol{\alpha} \boldsymbol{z}_T + u$ where $\boldsymbol{z}_T$ are control variables and $u$ is an error term. Using $m = e_T - w_H t_H - w_W t_W$ and expressions (6), (9) and (10), we obtain the childcare expenditure equation:

$$m = \begin{cases} m^* & \text{if } m^* \geq 0, \\ 0 & \text{otherwise.} \end{cases} \tag{11}$$

where

$$m^* = \boldsymbol{\alpha} \boldsymbol{z}_T + \gamma_W \sqrt{\theta_W w_W} + \gamma_H \sqrt{\theta_H w_H} + u \tag{12}$$

If $m = 0$ (i.e., the solution is at a corner), the parents' childcare time equations switch regime and the prices of parents' productive childcare time must satisfy the following condition:

$$\boldsymbol{\alpha} \boldsymbol{z}_T + \gamma_W \sqrt{w_W^*} + \gamma_H \sqrt{w_H^*} + u = 0, \tag{13}$$

with $w_H^* = \nu \theta_H w_H$ and $w_W^* = \nu \theta_W w_W$ for some function $\nu$, to guarantee that the childcare constraint is satisfied. This equation can be inverted to recover $\nu$ and the solution can be substituted into (9) and (10).

Finally, the error terms $(u, v_W, v_H)$ are assumed to be jointly normally distributed with mean zero and an unconstrained variance-covariance matrix. In our estimations, the control variables $\boldsymbol{z}_{T,H}$, $\boldsymbol{z}_{T,W}$ and $\boldsymbol{z}_T$ include a constant, the spouses' age, the spouses' education, the number of children, as well as a dummy for the presence of children under the age of 3.

### 3.2.2 The Full Leisure Demand Equations

To begin with, we note that $\varphi_I(t_I) = (1 - \theta_I)t_I$ in spouses' utility functions since $\theta_I$ is constant, as previously mentioned. As a result, the total leisure enjoyed by each spouse is defined as $L_I = l_I + (1 - \theta_I)t_I$. With a linear specification, the prices of productive and leisure-like (or labor-like) childcare time have a simple interpretation: the term $\theta_I t_I$ can be seen as the fraction or percentage of childcare time that is perceived as pure labor, while the term $(1 - \theta_I)t_I$ as the fraction that is perceived as pure leisure. The linear specification also guarantees that the utility profit functions $\pi_W$ and $\pi_H$ are always equal to zero and can therefore be omitted from the analysis.

As previously discussed, our empirical model employs full expenditure — denoted as $Y$ — rather than full income as the primary explanatory variable. The time cost of children is derived from the childcare technology and defined by expression (6), while the monetary cost is specified as:

$$e_M = \log\left(1 + \exp\left(\boldsymbol{\kappa}_M \boldsymbol{z}_M + \kappa_H w_H + \kappa_W w_W + \kappa_Y Y\right)\right) \times \sqrt{n} \tag{14}$$

where $\boldsymbol{\kappa}_M$, $\kappa_H$, $\kappa_W$ and $\kappa_Y$ are parameters, $\boldsymbol{z}_M$ includes a constant, the parents' education and age, and $n$ is the number of children. The transformation in (14) is commonly referred to as the softplus function in machine learning. This function ensures that monetary costs remain strictly positive, while offering a smooth and differentiable alternative to the simple maximum operator or the exponential function.

The remaining full expenditure $Y_R$ is equal to $Y$ for couples without children and to $Y - e_M - e_T$ for couples with children. For couples with and without children, the full leisure demand equations are specified either as:

(i) a quadratic function of remaining full expenditure:

$$L_H = \boldsymbol{a}_H \boldsymbol{z}_{L,H} + b_H w_H + c_H w_W + d_H Y_R + e_H Y_R^2 + \varepsilon_H \tag{15}$$

$$L_W = \boldsymbol{a}_W \boldsymbol{z}_{L,W} + b_W w_W + c_W w_H + d_W Y_R + e_W Y_R^2 + \varepsilon_W \tag{16}$$

where $\boldsymbol{a}_I$, $b_I$, $c_I$, $d_I$ and $e_I$ are parameters while $\boldsymbol{z}_{L,I}$ are control variables, and $\varepsilon_I$ is an error term, or as:

(ii) a quadratic function of remaining full expenditure with a demographic shift:

$$L_H = \boldsymbol{a}_H(n)\boldsymbol{z}_{L,H} + b_H(n)w_H + c_H(n)w_W + d_H Y_R + e_H Y_R^2 + \varepsilon_H \tag{17}$$

$$L_W = \boldsymbol{a}_W(n)\boldsymbol{z}_{L,W} + b_W(n)w_F + c_W(n)w_H + d_W Y_R + e_W Y_R^2 + \varepsilon_W \tag{18}$$

where $\boldsymbol{a}_I(n) = \boldsymbol{a}_I + \boldsymbol{a}'_I\delta(n)$, $b_I(n) = b_I + b'_I\delta(n)$ and $c_I(n) = c_I + c'_I\delta(n)$ are functions of $n$, $a'_I$, $b'_I$ and $c'_I$ are parameters and $\delta(n)$ is dummy variable equal to one if $n > 0$ and 0 otherwise. This specification allows the presence of children to shift the full leisure demand function, while leaving the slope of the Engel curves unchanged.

In our estimations, the control variables $\boldsymbol{z}_{L,H}$ and $\boldsymbol{z}_{L,W}$ include a constant, spouses' education and spouses' age. The error terms $(\varepsilon_W, \varepsilon_H)$ are assumed to have mean zero and an unconstrained variance-covariance matrix.

# 4 Estimation Method and Empirical Results

## 4.1 Estimation Method

In this subsection, we first present the two stage estimation method based on the Maximum Likelihood method and then the Three-Stage Least Square method.

While the five equations could be estimated simultaneously, we simplify the estimation procedure by taking advantage of the two-stage decision-making structure and estimate the system recursively. A benefit is that potential specification errors are not conflated across stages, allowing for a clean identification and estimation of childcare time prices. Moreover, the two stages rely on different samples: the first-stage is estimated using only couples with children, while the second-stage is estimated using the full sample of couples. In the first-stage, the three-equation system for the childcare technology, represented by (9)–(12), is estimated. In the data, a sub-

stantial proportion of parents report zero expenditure on paid childcare services, whereas the number of parents reporting zero hours of childcare time per week is negligible and may reflect measurement errors. We therefore assume that the two equations for parental childcare time are uncensored, whereas the equation for childcare expenditure is censored at zero. For tractability, we do not model the regime switch induced by zero childcare expenditure in the childcare-time equations, as doing so would introduce substantial additional nonlinearity. The resulting system is thus a mixed specification, combining linear and Tobit components, and is estimated by the Maximum Likelihood (ML) method. In the second-stage, the time cost and the value of both full leisure times are computed for all observations using $\widehat{\theta}_W$ and $\widehat{\theta}_H$ obtained in the first-stage. The two-equation system for spouses' full leisure demand, represented by (15)–(16) or (17)–(18), is then estimated by the Three-Stage Least Squares (3SLS) method. This estimation method accounts for the potential endogeneity of full expenditure and the time cost computed in the first-stage problem. Such endogeneity may indeed arise for several reasons. Most notably, measurement errors in working time and childcare time — and, by construction, in leisure time — are directly reflected in the computed values of full expenditure and childcare cost, respectively. To address this issue, we replace both potentially endogenous variables with predicted counterparts. Although full expenditure and the time cost enter the second-stage equations only through their sum, we construct their predicted values separately. The predicted time cost is directly obtained from the first-stage estimates by setting the corresponding error term equal to zero. Full expenditure, by contrast, is predicted using an auxiliary OLS regression on a constant, spouses' education, spouses' age, spouses' hourly wages, weekly nonlabor income, and its square, with weekly nonlabor income and its square serving as the excluded instruments. This auxiliary regression is run separately for couples with and without children. While nonlabor income is a standard source of exogenous variation in labor-supply and demand models, as emphasized for instance by Blundell and MaCurdy (1999), the predicted time cost provides an additional and more model-specific source of variation. It captures variation in the opportunity cost of childcare time implied by the first-stage time-allocation equations and therefore contributes directly to the predicted full cost measure used in the second-stage.

## 4.2 Empirical Results

### 4.2.1 Estimates of the Childcare Equations

In Table 4, the estimated parameters for the childcare time and expenditure equations are reported for four different samples or specifications. The estimates and standard errors in columns A are based on the sample described in Table 2 and those in columns B are based on a restricted sample constructed by dropping the 10 largest and 10 smallest observations for full expenditure and for spouses' hourly wages. This restriction aims at mitigating the potential influence of outliers, ensuring that estimates reflect general patterns rather than a few extreme cases. The estimates and standard errors in columns C are based on an extended sample which includes all couples without restriction on the number of children. The estimates and standard errors in columns D are based on the same sample as for columns A except that it uses the original PSID childcare time values rather than the adjusted ones. Overall, the results are similar across samples, although the parameter estimates are generally imprecise. The intercepts in the childcare-time equations are much larger in columns D because they absorb the adjustment constants $k_H$ and $k_W$. In columns C, where all couples are included regardless of the number of children, the estimated effects of the child-related variables naturally change, reflecting the broader variation in family size. These differences notwithstanding, most control variables are not statistically significant, with a few exceptions. In particular, childcare expenditure increases significantly with parental education and with the number of children, but childcare time displays no comparable pattern.

As expected, a higher price of productive childcare time, $w_H^*$ or $w_W^*$, is associated with less time spent on childcare and greater expenditure on paid childcare services. More precisely, the derivative of the father's childcare time equation with respect to the price of his productive childcare time is given by:

$$\frac{\partial t_H}{\partial w_H^*} = -\frac{1}{2}\gamma_H \left(w_H^*\right)^{-3/2} - \frac{1}{2}\gamma_{HW} \left(w_W^*\right)^{1/2} \left(w_H^*\right)^{-3/2}.$$

Using the estimates in column A and evaluating the derivative at the median of the wage

Table 4: ML Estimates of Childcare Time and Expenditure Equations

| | | A | | B | | C | | D | |
|---|---|---|---|---|---|---|---|---|---|
| Param. | Variable | Est. | StErr. | Est. | StErr. | Est. | StErr. | Est. | StErr. |
| | | | | | Childcare Expenditure Equation | | | | |
| $\alpha_0$ | Intercept | -99.28 | 28.71 | -100.91 | 34.90 | -136.65 | 25.19 | -98.37 | 28.71 |
| $\alpha_1$ | Husband's Educ. | 15.82 | 8.99 | 16.82 | 8.31 | 11.00 | 8.04 | 15.59 | 9.01 |
| $\alpha_2$ | Wife's Educ. | 15.16 | 9.41 | 5.32 | 8.73 | 12.04 | 8.53 | 15.34 | 9.41 |
| $\alpha_3$ | Husband's Age | 6.99 | 13.19 | 16.04 | 11.90 | -9.63 | 10.77 | 7.00 | 13.19 |
| $\alpha_4$ | Wife's Age | 19.56 | 13.77 | 8.18 | 12.51 | 31.09 | 11.02 | 19.31 | 13.76 |
| $\alpha_5$ | Child under 3 | 9.40 | 8.08 | 2.57 | 7.35 | 7.29 | 6.67 | 9.42 | 8.07 |
| $\alpha_6$ | Number of Children | 30.08 | 7.87 | 22.27 | 7.14 | -12.27 | 7.09 | 30.05 | 7.86 |
| | | | | | Father's Childcare Time Equation | | | | |
| $\beta_{H,0}$ | Intercept | 6.07 | 2.35 | 5.58 | 3.09 | 3.20 | 2.15 | 27.40 | 4.62 |
| $\beta_{H,1}$ | Husband's Educ. | -0.53 | 1.13 | 0.04 | 1.19 | 0.24 | 0.98 | -0.70 | 2.18 |
| $\beta_{H,2}$ | Wife's Educ. | 0.01 | 1.18 | 0.22 | 1.25 | 0.81 | 1.03 | 0.11 | 2.26 |
| $\beta_{H,3}$ | Husband's Age | 0.25 | 1.64 | 0.46 | 1.70 | -2.14 | 1.30 | 0.07 | 3.15 |
| $\beta_{H,4}$ | Wife's Age | -1.32 | 1.71 | -1.65 | 1.78 | 0.44 | 1.34 | -2.79 | 3.29 |
| $\beta_{H,5}$ | Child under 3 | 0.31 | 1.01 | 0.97 | 1.05 | 0.63 | 0.82 | 1.63 | 1.93 |
| $\beta_{H,6}$ | Number of Children | -0.56 | 0.97 | 0.09 | 1.01 | 0.08 | 0.84 | -0.13 | 1.87 |
| | | | | | Mother's Childcare Time Equation | | | | |
| $\beta_{W,0}$ | Intercept | 12.95 | 2.52 | 8.14 | 4.79 | 14.39 | 2.36 | 44.72 | 4.10 |
| $\beta_{W,1}$ | Husband's Educ. | -2.53 | 1.54 | -1.62 | 1.58 | -1.08 | 1.45 | -3.96 | 2.51 |
| $\beta_{W,2}$ | Wife's Educ. | -0.05 | 1.56 | 0.19 | 1.67 | 0.40 | 1.51 | 0.72 | 2.56 |
| $\beta_{W,3}$ | Husband's Age | -0.33 | 2.24 | 1.29 | 2.28 | -2.51 | 1.94 | -0.88 | 3.67 |
| $\beta_{W,4}$ | Wife's Age | -1.73 | 2.32 | -2.89 | 2.37 | 0.24 | 1.98 | -2.92 | 3.80 |
| $\beta_{W,5}$ | Child under 3 | 0.40 | 1.38 | 1.08 | 1.40 | 1.39 | 1.22 | 0.67 | 2.26 |
| $\beta_{W,6}$ | Number of Children | -1.17 | 1.33 | -0.37 | 1.35 | 2.11 | 1.25 | -1.90 | 2.17 |
| | | | | | Wage Coefficients | | | | |
| $\gamma_H$ | | 20.51 | 6.67 | 22.44 | 7.41 | 28.39 | 5.90 | 27.50 | 9.85 |
| $\gamma_W$ | | 27.32 | 6.38 | 36.77 | 9.44 | 28.72 | 5.96 | 31.45 | 8.59 |
| $\gamma_{W,H}$ | | -3.26 | 2.46 | -4.84 | 2.90 | -3.63 | 2.39 | -4.07 | 4.61 |
| | | | | | Price-Scales Coefficients | | | | |
| $\theta_H$ | | 0.35 | 0.21 | 0.28 | 0.20 | 0.34 | 0.13 | 0.20 | 0.12 |
| $\theta_W$ | | 0.44 | 0.26 | 0.30 | 0.14 | 0.42 | 0.24 | 0.32 | 0.24 |

Notes: All demographic variables are standardized. A = benchmark sample (463 observations); B = sample excluding extreme values (413 observations); C = sample including households with more than two children (681 observations); D = sample using the original PSID childcare time measures (463 observations).

distributions, this corresponds to a marginal effect of approximately $-0.18$ (StdErr = 0.11). Thus, a one-dollar increase in the price of productive childcare time is associated with about 0.18 fewer hours of childcare per week or approximately 11 minutes. The effect is moderate at the median but strongly nonlinear. At the first decile of the men's wage distribution, holding the women's wage at its median, the marginal effect is about $-0.65$ hours (StdErr = 0.38) whereas at the ninth decile it falls to about -0.05 (StdErr=0.03). Similarly, the derivative with respect to the mother's price of productive childcare time is given by:

$$\frac{\partial t_H}{\partial w_W^*} = \frac{1}{2}\gamma_{HW}\left(w_H^*\right)^{-1/2}\left(w_W^*\right)^{-1/2}$$

Evaluated at the median of the wage distributions, the marginal effect is equal to $-0.17$ (StdErr = 0.13) or approximately 10 minutes. The negative cross-wage effect implies that the two parental time inputs can be seen as Hicks–Allen complements. Finally, the derivative of the mother's childcare time equation with respect to her price of productive childcare time is equal to $-0.27$ hours (StdErr = 0.09) or approximately 16 minutes while the derivative with respect to the father's price of time is, by symmetry, also equal to $-0.17$ (StdErr = 0.13).[27] The fact that the parameters $\gamma_{WH}$ is negative implies that the cost function is not globally consistent. Unreported results indicate that the error terms in the parents' childcare time equations are positively correlated. The mother's childcare time error term is also positively correlated with that of childcare expenditures, while the corresponding correlation for the father is insignificant.

The most important parameters in the model are the price-scale coefficients $\theta_H$ and $\theta_W$. Under the linear specification of process benefits, these parameters can be interpreted as the proportion

[27]The number of studies that explicitly examine the relationship between childcare time and hourly wages is very limited. They generally do not find significant effects of hourly wages or confirm the negative effect we obtain (Bloemen and Stancanelli, 2014, for France; Kalenkoski, Ribar, Stratton, 2009, for U.K.; Hallberg and Klevmarken, 2003, and Gustafsson and Kjullin, 1994, for Sweden). Among the very few available studies based on U.S. data, Kimmel and Connelly (2007) present the surprising finding, with the American Time Use Survey (ATUS), that an increase in wage rates is associated with an increase in time spent on childcare. They interpret this as an income-effect mechanism: as wages rise, households demand higher-quality childcare activities. Note that their sample includes both couples and single-parent households, with multiple and older children.

of childcare time that is perceived as pure labor by the mother and the father, respectively. Using the benchmark sample, our results suggest that about 44% of the time mothers spend caring for their children is treated as labor, while the remainder resembles leisure, while this proportion is lower for fathers and equal to about 35%. Using the sample which trims extreme values (columns B) or the original PSID childcare time measures (columns D), the coefficients are somewhat lower but not significantly different. Overall, the estimated price-scale coefficients are broadly similar in magnitude to the back-of-the-envelope approximations obtained from preliminary OLS specifications.

To assess robustness, we also re-estimate models with a more parsimonious set of covariates; the results change little even if the price-scale coefficients are slightly larger. In addition, we fit a more flexible cost function that includes a quadratic in hourly wages; the added terms are statistically indistinguishable from zero at the 5% level. Finally, following Proposition 3, we test the exclusion of full expenditure from the childcare equations using a likelihood-ratio (LR) test. Since full expenditure may be endogenous, we replace it with its natural instrument, non-labor income, which is also used as an instrument in the estimation of the full leisure demand equations below. We do not reject the joint null hypothesis that the three coefficients on non-labor income in the childcare equations are zero at conventional significance levels. For the extended sample (columns C), however, the p-value is only slightly above 0.05. Overall, these diagnostics provide no strong evidence against the exclusion restriction.

That said, the estimated price-scale coefficients might be subject to downward bias for at least two reasons. First, the demand for formal childcare services may be rationed — either because the couple lives in a region with limited availability of childcare facilities, or because available services do not match the household's preferences in terms of quality, schedule, or accessibility. Second, the actual cost of using formal childcare services may be higher than what is reported in the PSID, as it may involve fixed, unobserved costs — such as commuting to the care provider, or the time and effort required to coordinate logistics, including scheduling and drop-off/pick-up arrangements. These factors could attenuate the sensitivity of the demand for childcare services and lead to an underestimation of price-scale coefficients.

Despite these potential limitations, the estimated price-scale coefficients appear plausible and provide a solid basis for computing both spouses' full leisure time and the time cost of children.

### 4.2.2 Estimates of the Full Leisure Demand Equations

This subsection sequentially presents the parameter estimates of Engel curves and the monetary cost of children. In Table 5, the columns A display the estimates and standard errors obtained from the subsample of childless couples — and thus does not involve the estimation of the monetary cost of children. The columns B and C correspond, respectively, to the estimates and standard errors of the standard Quadratic Engel curve specification and of the Quadratic, Demographic-Shift Engel curve specification. The columns D correspond to the same specification as that in columns B with an alternative definition full leisure. In this definition, the time devoted to domestic chores is subtracted from the previous measure of leisure. The bottom panel of the table reports the estimated parameters of the monetary cost function. Without regularization, these parameters are highly unstable and can take implausibly large values. We therefore use a ridge penalty to stabilize the estimates. The choice of this parameter and its implications are discussed in detail below.

Using the childless sample, we find a positive but weakly significant relationship between full expenditure and full leisure, defined as non-working time. A Wald test does not reject the null hypothesis of a flat Engel-curve slope in either equation. Including couples with children leaves the estimates essentially unchanged but reduces the standard errors considerably, owing to the larger sample, greater variation in both full leisure and full expenditure, and the strong instrument generated by the fitted time cost. Intuitively, the large differences in full leisure between childless couples and parents is attributed in the model to the full cost of children, a fraction of which is represented by the time cost. The parameters associated to Engel curves in columns B are generally significantly different from zero, with the notable exception of the quadratic terms. In particular, a \$1,000 increase in full expenditure is associated (ignoring quadratic terms to simplify) with an additional 2.8 hours of full leisure for men and and 3.8 hours for women, while higher hourly wages are associated with less full leisure for both spouses. The columns C of Table 5 report estimates and standard errors for the Quadratic, Demographic-Shift

Table 5: 3SLS Estimates of Full Leisure Demand Equations

| | | A | | B | | C | | D | |
|---|---|---|---|---|---|---|---|---|---|
| Param. | Variables | Est. | St.Err. | Est. | St.Err. | Est. | St.Err. | Est. | St.Err. |
| | | | | | Husband' Engel Curve | | | | |
| $a_{H,0}$ | Intercept | 74.95 | 12.23 | 72.80 | 2.64 | 72.33 | 2.66 | 64.68 | 2.77 |
| $a_{H,1}$ | Husband's Educ. | -1.03 | 2.61 | -1.77 | 0.63 | -1.10 | 0.87 | -2.14 | 0.68 |
| $a_{H,2}$ | Wife's Educ. | 0.66 | 2.37 | 0.69 | 0.62 | 0.52 | 0.78 | 0.14 | 0.68 |
| $a_{H,3}$ | Husband's Age | -0.53 | 3.60 | -0.63 | 0.96 | -0.66 | 1.20 | -0.80 | 1.05 |
| $a_{H,4}$ | Wife's Age | 0.07 | 3.52 | 0.17 | 0.97 | -0.04 | 1.17 | 0.24 | 1.06 |
| $b_H$ | Husband's Wage | 0.04 | 0.80 | -0.26 | 0.14 | -0.16 | 0.17 | -0.31 | 0.14 |
| $c_H$ | Wife's Wage | -0.10 | 0.72 | -0.28 | 0.13 | -0.28 | 0.15 | -0.37 | 0.13 |
| $d_H$ | Full Expend. | 0.58 | 7.86 | 2.79 | 1.35 | 2.66 | 1.48 | 3.76 | 1.39 |
| $e_H$ | Sq. Full Expend. | -0.00 | 0.13 | 0.01 | 0.03 | -0.01 | 0.03 | -0.01 | 0.03 |
| | | | | | Husband's Demographic Shift | | | | |
| $a'_{H,0}$ | Intercept | . | . | . | . | -0.24 | 3.48 | . | . |
| $a'_{H,1}$ | Husband's Educ. | . | . | . | . | -0.98 | 1.35 | . | . |
| $a'_{H,2}$ | Wife's Educ. | . | . | . | . | -0.40 | 1.38 | . | . |
| $a'_{H,3}$ | Husband's Age | . | . | . | . | -0.51 | 1.82 | . | . |
| $a'_{H,4}$ | Wife's Age | . | . | . | . | 0.81 | 1.86 | . | . |
| $b'_H$ | Husband's Wage | . | . | . | . | -0.12 | 0.06 | . | . |
| $c'_H$ | Wife's Wage | . | . | . | . | 0.10 | 0.07 | . | . |
| | | | | | Wife's Engel Curve | | | | |
| $a_{W,0}$ | Intercept | 80.73 | 13.12 | 80.32 | 3.30 | 80.55 | 3.15 | 65.11 | 4.40 |
| $a_{W,1}$ | Husband's Educ. | -0.64 | 2.80 | -0.44 | 0.74 | -0.70 | 0.95 | 0.41 | 0.93 |
| $a_{W,2}$ | Wife's Educ. | -0.72 | 2.55 | -1.02 | 0.72 | -0.68 | 0.86 | -1.69 | 0.92 |
| $a_{W,3}$ | Husband's Age | -0.56 | 3.87 | -1.33 | 1.10 | -0.49 | 1.32 | 0.22 | 1.39 |
| $a_{W,4}$ | Wife's Age | 0.27 | 3.78 | 0.66 | 1.10 | 0.29 | 1.29 | -1.33 | 1.38 |
| $b_W$ | Husband's Wage | -0.46 | 0.86 | -0.41 | 0.18 | -0.46 | 0.18 | -0.73 | 0.24 |
| $c_W$ | Wife's Wage | -0.56 | 0.77 | -0.46 | 0.17 | -0.34 | 0.20 | -0.70 | 0.22 |
| $d_W$ | Full Expend. | 4.29 | 8.44 | 3.82 | 1.80 | 3.54 | 1.84 | 7.00 | 2.46 |
| $e_W$ | Sq. Full Expend. | 0.03 | 0.14 | 0.03 | 0.04 | 0.01 | 0.04 | 0.01 | 0.05 |
| | | | | | Wife's Demographic Shift | | | | |
| $a'_{W,0}$ | Intercept | . | . | . | . | -1.84 | 4.24 | . | . |
| $a'_{W,1}$ | Husband's Educ. | . | . | . | . | 0.81 | 1.61 | . | . |
| $a'_{W,2}$ | Wife's Educ. | . | . | . | . | -1.39 | 1.65 | . | . |
| $a'_{W,3}$ | Husband's Age | . | . | . | . | -2.65 | 2.02 | . | . |
| $a'_{W,4}$ | Wife's Age | . | . | . | . | 0.99 | 2.07 | . | . |
| $b'_W$ | Husband's Wage | . | . | . | . | 0.08 | 0.08 | . | . |
| $c'_W$ | Wife's Wage | . | . | . | . | 0.01 | 0.07 | . | . |
| | | | | | Monetary Cost | | | | |
| $\kappa_0$ | Intercept | . | . | 5.36 | 2.92 | 4.54 | 2.24 | 3.87 | 1.99 |
| $\kappa_1$ | Husband's Educ. | . | . | -0.24 | 0.33 | -0.16 | 0.38 | -0.17 | 0.22 |
| $\kappa_2$ | Wife's Educ. | . | . | 0.45 | 0.37 | 0.06 | 0.43 | 0.08 | 0.23 |
| $\kappa_3$ | Husband's Age | . | . | 0.06 | 0.47 | -0.26 | 0.20 | 0.20 | 0.32 |
| $\kappa_4$ | Wife's Age | . | . | 0.01 | 0.51 | 0.02 | 0.21 | -0.10 | 0.35 |
| $\kappa_H$ | Husband's Wage | . | . | -0.08 | 0.12 | -0.12 | 0.05 | -0.05 | 0.09 |
| $\kappa_W$ | Wife's Wage | . | . | 0.00 | 0.12 | 0.02 | 0.02 | -0.03 | 0.09 |
| $\kappa_Y$ | Full Expend. | . | . | -0.55 | 1.24 | -0.32 | 0.16 | -0.24 | 0.93 |

Notes: All demographic variables are standardized. A = childless sample, B = Quadratic Engel curve specification, C = Quadratic, Demographic-Shift Engel curve specification, D = alternative definition of leisure.

Engel curve specification. The results remain very similar overall, although standard errors are somewhat larger, and none of the additional demographic-shift terms is statistically significant at conventional levels, either individually or jointly according to a Wald test. Nevertheless, the signs of the estimated shifts are informative: both intercept shifts are negative, suggesting that part of the effect of children may be absorbed by these shifts rather than by the monetary cost function. The estimates and standard errors based on an alternative measure of leisure are reported in columns D. The results are very similar to those in columns B, although the estimated effects of full expenditure are slightly larger.

By contrast, the parameters of the monetary cost function are much more difficult to estimate and tend to be highly unstable. To address this issue, we introduce a ridge penalty of the form $\zeta \times (\kappa_1^2 + ... + \kappa_Y^2)$ applied to the slope coefficients but not to the intercept, so that excessively large values are penalized. The regularization parameter $\zeta > 0$ controls the strength of the regularization and is set to a small value, with the sole purpose of stabilizing the estimation while limiting the amount of shrinkage. With this regularization, the resulting estimates are stable but not statistically significant at conventional levels. In any case, since ridge regularization introduces shrinkage and affects conventional standard errors, inference on these coefficients should be interpreted with caution. This caveat applies with particular force to the estimates reported in columns C. Since the quadratic terms in the Engel curves are themselves insignificant, identification relies on the nonlinearity of the monetary cost function. Indeed, replacing this function with a linear specification leads to a dramatic increase in standard errors. For these reasons, the empirical interpretation should focus primarily on the average level of the monetary cost of children rather than on the estimated effects of individual covariates.

One final remark is in order. Because the model is overidentified and the monetary cost parameters can be identified from either spouse's full-leisure equation alone, we also estimate a specification allowing these parameters to differ across spouses. A distance test based on the standard Quadratic Engel curve specification yields a chi-squared statistic of 12.19 with fourteen degrees of freedom, indicating that allowing for spouse-specific monetary cost parameters does not significantly improve model fit. We also assess the exclusion restriction on nonlabor income by re-estimating the standard Quadratic Engel curve specification with nonlabor income and

Table 6: Descriptive Statistics for the Distribution of the Weekly Time and Monetary Cost of Children (in Thousands of Dollars per Week)

| | Children | count | mean | std | min | 25% | 50% | 75% | max |
|---|---|---|---|---|---|---|---|---|---|
| | | Weekly time cost in thousands dollars | | | | | | | |
| A | 1 | 205.00 | 0.43 | 0.64 | 0.00 | 0.08 | 0.23 | 0.52 | 5.14 |
| | 2 | 258.00 | 0.40 | 0.50 | 0.00 | 0.10 | 0.25 | 0.48 | 3.63 |
| B | 1 | 183.00 | 0.29 | 0.36 | 0.00 | 0.07 | 0.17 | 0.36 | 2.58 |
| | 2 | 230.00 | 0.29 | 0.36 | 0.00 | 0.08 | 0.20 | 0.36 | 2.87 |
| C | 1 | 205.00 | 0.42 | 0.61 | 0.00 | 0.08 | 0.22 | 0.51 | 4.94 |
| | 2 | 258.00 | 0.38 | 0.48 | 0.00 | 0.09 | 0.24 | 0.47 | 3.50 |
| D | 1 | 205.00 | 0.81 | 0.90 | 0.00 | 0.33 | 0.57 | 0.97 | 8.07 |
| | 2 | 258.00 | 0.76 | 0.73 | 0.00 | 0.32 | 0.56 | 0.94 | 6.15 |
| A* | 1 | 205.00 | 0.93 | 1.54 | 0.00 | 0.14 | 0.35 | 1.08 | 11.47 |
| | 2 | 258.00 | 0.80 | 1.21 | 0.00 | 0.15 | 0.39 | 0.90 | 9.83 |
| | | Weekly monetary cost in thousands dollars | | | | | | | |
| B | 1 | 203.00 | 0.92 | 0.96 | 0.00 | 0.08 | 0.62 | 1.52 | 3.66 |
| | 2 | 252.00 | 1.30 | 1.31 | 0.00 | 0.23 | 0.85 | 1.98 | 6.17 |
| C | 1 | 203.00 | 1.05 | 1.03 | 0.00 | 0.12 | 0.80 | 1.75 | 4.17 |
| | 2 | 252.00 | 1.49 | 1.42 | 0.00 | 0.26 | 1.11 | 2.22 | 6.08 |
| D | 1 | 203.00 | 1.01 | 0.80 | 0.00 | 0.27 | 0.89 | 1.54 | 3.06 |
| | 2 | 252.00 | 1.52 | 1.12 | 0.00 | 0.60 | 1.34 | 2.23 | 4.43 |

Notes: Computations do not use PSID sample weights. For time cost, A = benchmark sample, B = sample without extreme values, C = sample with households with more than 2 children, D = sample with original PSID childcare time measures, A* = benchmark sample with childcare time valued at observed wages. For monetary cost, B = Quadratic Engel curve specification, C = Quadratic, Demographic-Shift Engel curve specification, D = alternative definition of leisure.

its square entering directly into both spouses' full-leisure equations. A Wald test of the joint null that the four additional coefficients are zero yields a chi-squared statistic of 0.876 with four degrees of freedom. The null hypothesis is not rejected, providing support for the exclusion restriction.

### 4.2.3 Estimates of the time and monetary cost of children

The characteristics of the distribution of time and monetary costs, across the different specifications and for one and two children compositions, are reported in Table 6. The median time cost for a single child is sizable — about $ 230 per week (row A), $ 170 when trimming extreme values (row B), $ 220 when using the extended sample with more than two children (row C) and $ 570 when using the original PSID childcare time measure (row D) — representing between 7% and 25% of total spousal labor income. The differences between the first three specifications are mainly driven by variations in the price of childcare time, evaluated by the estimation of childcare equations. The fourth specification relies on a time measure that includes passive childcare (e.g., supervisory time) which likely leads to an overstatement of the true time cost. By comparison, when parental childcare time is instead valued at observed wages (row A$^*$), the median time cost of a single child rises to about $350 per week, roughly 50% above the benchmark estimate based on adjusted shadow prices. The density function of time costs under the benchmark specification (row A), pooling the sample of couples with one and two children, is shown in Figure 1. The distribution is markedly right-skewed, with a long upper tail, so mean values exceed medians substantially and are likely less representative of a typical household.

Finally, the monetary cost for a single child, evaluated at its sample median, is estimated at around $620 per week for the standard Quadratic Engel curve specification (row B), $800 per week for the Quadratic, Demographic-Shift Engel curve specification (row C), and $890 per week for the sample with the alternative definition of leisure (row D). The density function of monetary costs under the Quadratic Engel curve specification (row B) is shown in Figure 2 and is likewise distinctly right-skewed.[28]

[28]Standard errors can be computed for costs evaluated at a specific point of the distribution using the delta method. For each component, we evaluate the cost at the median of the estimated distribution for one-child households and compute the standard error conditional on the characteristics of the corresponding household. The resulting time and monetary costs are about $232 (StdErr = 139) and $616 (StdErr = 420), respectively.

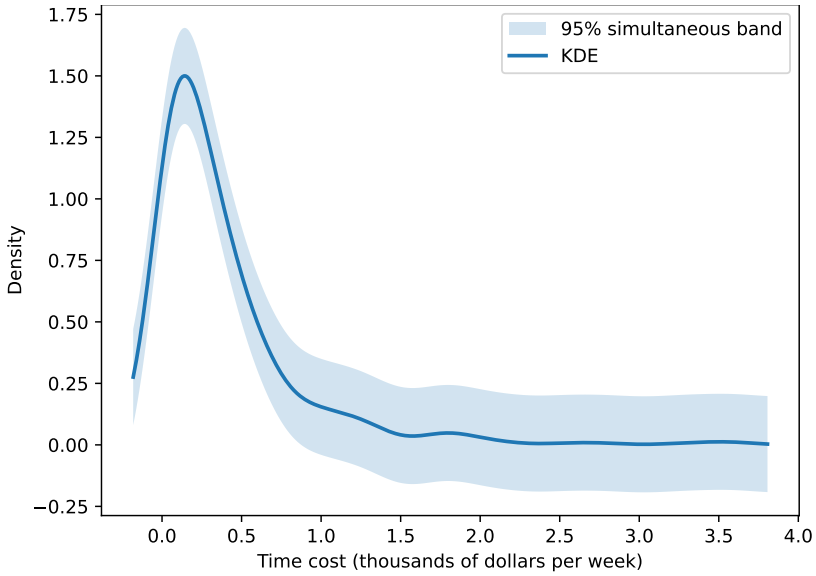



Figure 1: Distribution of Time Costs (in thousands of dollars per week)

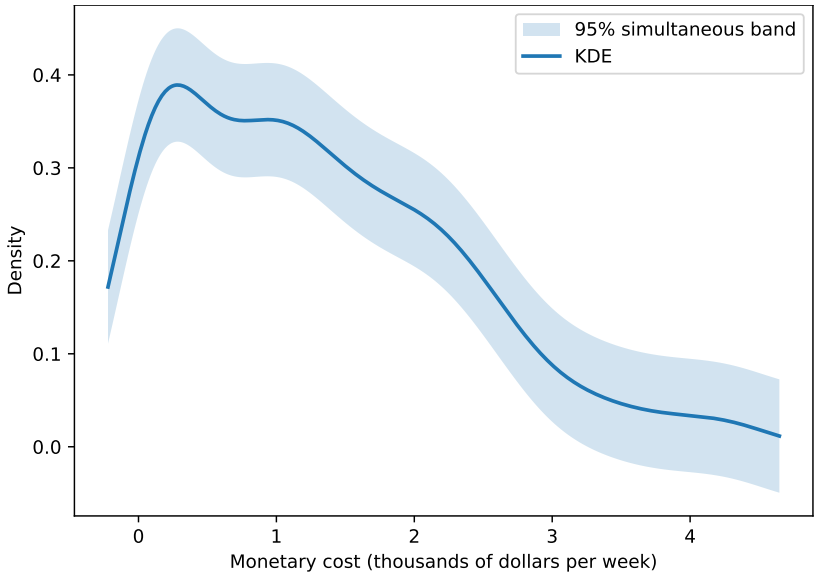



Figure 2: Distribution of Monetary Costs (in thousands of dollars per week)

Notes: Computations do not use PSID sample weights. The solid lines report kernel density estimates of the distributions of estimated costs. The shaded areas are 95% simultaneous bootstrap confidence bands for the densities, based on 1,000 resamples of households with model parameters held fixed. They therefore do not incorporate the uncertainty of parameter estimation.

# 5 Conclusion

In this paper, we have developed a static model to evaluate the full cost of children, including both a monetary component and a time component. A crucial point of our study lies in recognizing that the price of time used to assess the time cost of children is not necessarily equivalent to the parents' wage rate. Instead, it depends on how parents perceive their time spent with their children — whether they consider it closer to leisure or to labor. To the best of our knowledge, this is the first study to explicitly make this distinction and to provide estimates of the full cost of children that account for this nuanced valuation of time.

Estimating the model requires detailed information on both the time parents devote to their children and their spending on external childcare. We implement it using the 2019 PSID, which provides the necessary data. Recovering the full cost of children raises two challenges, though. First, the full leisure demand equations are only weakly responsive to changes in full expenditure — the associated Engel curves are relatively flat — so identification based on leisure alone is potentially weak. Importantly, our procedure first recovers the time cost and then the full cost of children, with the monetary cost obtained residually as the difference between the two. This sequence is not essential, however. Alternative implementations could rely on adult goods with steeper Engel curves, such as clothing, either to recover the monetary component — which would then be added to the independently identified time component to obtain the full cost — or, under suitable additional conditions, to identify the full cost directly. Exploring these alternatives would therefore require a substantially different empirical strategy and is left for future research. Second, full expenditure must be instrumented, and credible instruments are scarce. Our theoretical framework nonetheless delivers a strong instrument — the fitted value of the child time cost obtained from the childcare equations — although it is model-dependent.

Notwithstanding these caveats, our baseline results indicate that one hour of the mother's productive childcare time is priced at approximately 44% of her hourly wage, and one hour of the father's at 35%. The prices of productive childcare time differ from current hourly wages because a fraction of childcare time can be assimilated to pure leisure. The resulting time cost of children is substantial: for a single child, its median is about $230 per week. The corresponding median

monetary cost is approximately $620 per week, implying a total cost in excess of $800 per week, well above typical estimates in the literature. These figures, moreover, capture only short-run costs. They exclude longer-term consequences such as career interruptions and reduced human capital accumulation, particularly for mothers, so the overall economic cost of children may be considerably larger.

Beyond these estimates, the methodology developed in this paper has several potential applications. By operationalizing Reid's (1934) third-party criterion, it provides a measure of the value of childcare services produced within the household that could otherwise be supplied by the market, and may therefore contribute to household satellite accounts and broader measures of national production. In particular, it offers an alternative to the common practice of valuing all parental childcare time at the market wage, which may substantially overstate the value of household childcare production when parental time also generates direct utility. For practical applications in which the full model cannot be estimated, our results suggest that valuing parental childcare time at about 30–50 percent of the wage may provide a useful rule of thumb and a more realistic benchmark than using the full wage rate.

The resulting measure of the full cost of children can also be used in welfare analysis, for instance to construct equivalence scales or assess poverty and inequality after accounting for both monetary expenditures and parental time costs. More specifically, the structural framework can support the evaluation of policies that alter the relative cost of parental and market-provided childcare, such as childcare subsidies, tax credits, or public daycare provision. Such policies may affect not only household expenditures on childcare but also the allocation of care between parents and the market. By distinguishing the monetary and time components of the cost of children, the framework makes it possible to evaluate these adjustments without mechanically valuing every change in parental childcare time at the wage rate.

# Appendix A : Proofs

**Proof of Proposition 1.**

The first order conditions of the optimization problem $\bar{\text{P}}$ are:

$$l_W : \phi_W \frac{\partial U_W}{\partial l_W} - \lambda_1 w_W = 0 \tag{A.1}$$

$$t_W : \phi_W \frac{\partial U_W}{\partial t_W} - \lambda_1 w_W - \lambda_2 \frac{\partial g}{\partial t_W} = 0 \tag{A.2}$$

$$c_W : \phi_W \frac{\partial U_W}{\partial c_W} - \lambda_1 = 0 \tag{A.3}$$

$$l_H : \phi_H \frac{\partial U_H}{\partial l_H} - \lambda_1 w_H = 0 \tag{A.4}$$

$$t_H : \phi_H \frac{\partial U_H}{\partial t_H} - \lambda_1 w_H - \lambda_2 \frac{\partial g}{\partial t_H} = 0 \tag{A.5}$$

$$c_H : \phi_H \frac{\partial U_H}{\partial c_H} - \lambda_1 = 0 \tag{A.6}$$

$$c_K : \phi_H \frac{\partial U_H}{\partial c_K} + \phi_W \frac{\partial U_W}{\partial c_K} - \lambda_1 = 0 \tag{A.7}$$

$$m : \lambda_2 - \lambda_1 + \mu = 0 \tag{A.8}$$

where $\lambda_1$ and $\lambda_2$ are the Lagrange multipliers for the budget constraint (3) and the childcare technology constraint (1), respectively, and $\mu$ is the Kuhn-Tucker multiplier for the nonnegativity constraint (2), with the following (dual and complementary slackness) conditions:

$$\mu m = 0 \text{ and } \mu \geq 0.$$

**Part 1.** Define $\lambda = \lambda_2$ and, from (A.8), write $\lambda + \mu = \lambda_1$. Then, from (A.2), we obtain:

$$w_W \cdot \left(1 - \frac{\phi_W}{w_W (\lambda + \mu)} \frac{\partial U_W}{\partial t_W}\right) + \frac{\lambda}{\mu + \lambda} \cdot \frac{\partial g}{\partial t_W} = 0$$

Define $\nu = 1 + \mu/\lambda$ and

$$\theta_W = 1 - \frac{\phi_W}{\lambda \nu w_W} \frac{\partial U_W}{\partial t_W},$$

so that we obtain:

$$w_W \cdot \nu \cdot \theta_W + \frac{\partial g}{\partial t_W} = 0. \tag{A.9}$$

Using (A.1)

$$\theta_W = 1 - \frac{\phi_W}{\lambda \nu w_W} \frac{\partial U_W}{\partial t_W} = 1 - \frac{\partial U_W / \partial t_W}{\partial U_W / \partial l_W} > 0,$$

from the assumptions on utility functions. Similarly, from (A.5), we obtain:

$$w_H \cdot \nu \cdot \theta_H + \frac{\partial g}{\partial t_H} = 0. \tag{A.10}$$

This system of two equations (A.9)–(A.10) constitutes the first-order conditions of the optimization problem $P_0$:

$$t_W : \theta_W w_W + \frac{\partial g}{\partial t_W}(1 - \zeta) = 0$$
$$t_H : \theta_H w_H + \frac{\partial g}{\partial t_H}(1 - \zeta) = 0$$

where $\zeta$ is the Kuhn-Tucker multiplier associated with the constraint $g(t_W, t_H, n) \geq 0$, with

$$1 - \zeta = \frac{1}{\nu} = \frac{1}{\lambda + \mu}.$$

**Part 2.** We write: $\lambda_I = \lambda_1/\phi_I$. From (A.1)–(A.3) or (A.4)–(A.6), we obtain:

$$\frac{\partial U_I}{\partial l_I} - \lambda_I w_I = 0, \tag{A.11}$$

$$\frac{\partial U_I}{\partial t_I} - \lambda_I \left(1 - \overline{\theta}_I\right) w_I = 0, \tag{A.12}$$

$$\frac{\partial U_I}{\partial c_I} - \lambda_I = 0, \tag{A.13}$$

with $I = W$ and $H$. Conditional on $c_K$, this system of three equations (A.11)–(A.13) constitutes the first-order conditions of the optimization problem $\mathrm{P}_I$, where $\lambda_I$ is the Lagrange Multiplier of the budget constraint.

**Part 3.** By the envelope theorem, the first-order conditions of Problems $\mathrm{P}_W$ and $\mathrm{P}_H$ imply:

$$\frac{\partial V_I}{\partial \kappa_I} = \lambda_I, \qquad \frac{\partial V_I}{\partial c_K} = \frac{\partial U_I}{\partial c_K},$$

with $I = H, W$. Since $\lambda_I = \lambda_1/\phi_I$, it follows that:

$$\phi_W \frac{\partial V_W}{\partial \kappa_W} = \lambda_1, \qquad \phi_H \frac{\partial V_H}{\partial \kappa_H} = \lambda_1,$$

and, from (A.7), it follows that:

$$\phi_H \frac{\partial V_H}{\partial c_K} + \phi_W \frac{\partial V_W}{\partial c_K} - \lambda_1 = 0$$

Hence, the first-order conditions of the optimization problem $\mathrm{P}_2$ are satisfied.

**Remark.** The proof above assumes that the time-feasibility constraints are slack, so that both spouses supply a strictly positive amount of market work. Suppose instead that spouse $I$ does not work in the market. Then

$$l_I + t_I = T.$$

and the corresponding time-feasibility constraint is binding. Let $\eta_I \geq 0$ denote its Kuhn–Tucker multiplier. The first-order conditions with respect to $l_I$ and $t_I$ become:

$$l_I : \phi_I \frac{\partial U_I}{\partial l_I} - \lambda_1 w_I - \eta_I = 0 \tag{A.14}$$

$$t_I : \phi_I \frac{\partial U_I}{\partial t_I} - \lambda_1 w_I - \lambda_2 \frac{\partial g}{\partial t_I} - \eta_I = 0 \tag{A.15}$$

Define the shadow price of spouse $I$'s time by

$$w_I^s = w_I + \frac{\eta_I}{\lambda_1}$$

The preceding conditions can then be rewritten as

$$\phi_I \frac{\partial U_I}{\partial l_I} - \lambda_1 w_I^s = 0 \tag{A.16}$$

$$\phi_I \frac{\partial U_I}{\partial t_I} - \lambda_1 w_I^s - \lambda_2 \frac{\partial g}{\partial t_I} = 0 \tag{A.17}$$

Hence, all the arguments used in the proof continue to apply after replacing the observed wage $w_I$ by the shadow wage $w_I^s$. In particular, the first-order condition of the childcare-cost problem becomes:

$$w_I^s \cdot \nu \cdot \theta_I + \frac{\partial g}{\partial t_I} = 0.$$

To preserve the equivalence with the original resource constraint, the use of shadow wages must be accompanied by a corresponding adjustment in full income. Define:

$$Y^s = Y + T \cdot (w_I^s - w_I)$$

so that the decentralized problems retain the same form after replacing $w_I$ by $w_I^s$ and $Y$ by $Y^s$. □

**Proof of Proposition 2.**

The proof follows in two stages.

**Part 1.** Suppose first that $(w_W, w_H, Y, n)$ is such that $m^*(w_W, w_H, Y, n) > 0$. Given that the childcare constraint is binding at equilibrium, we have:

$$m^*(w_W, w_H, Y, n) = g(t_W^*(w_W, w_H, Y, n), t_H^*(w_W, w_H, Y, n), n), \tag{A.18}$$

Consider a neighborhood of $(w_W, w_H, Y, n)$ in which $m^*(w_W, w_H, Y, n) > 0$. Differentiating both sides of this equation with respect to $w_W$ and $w_H$, holding $Y$ and $n$ fixed, gives:

$$\frac{\partial m^*}{\partial w_W} = \frac{\partial g}{\partial t_W} \cdot \frac{\partial t_W^*}{\partial w_W} + \frac{\partial g}{\partial t_H} \cdot \frac{\partial t_H^*}{\partial w_W}, \tag{A.19}$$

$$\frac{\partial m^*}{\partial w_H} = \frac{\partial g}{\partial t_W} \cdot \frac{\partial t_W^*}{\partial w_H} + \frac{\partial g}{\partial t_H} \cdot \frac{\partial t_H^*}{\partial w_H}. \tag{A.20}$$

Equivalentely,

$$\nabla m = \mathbf{J}_t \cdot \nabla g$$

where

$$\nabla m = \begin{pmatrix} \dfrac{\partial m^*}{\partial w_W} \\ \dfrac{\partial m^*}{\partial w_H} \end{pmatrix}, \qquad \nabla g = \begin{pmatrix} \dfrac{\partial g}{\partial t_W} \\ \dfrac{\partial g}{\partial t_H} \end{pmatrix}, \qquad \mathbf{J}_t = \begin{pmatrix} \dfrac{\partial t_W^*}{\partial w_W} & \dfrac{\partial t_H^*}{\partial w_W} \\ \dfrac{\partial t_W^*}{\partial w_H} & \dfrac{\partial t_H^*}{\partial w_H} \end{pmatrix}. \tag{A.21}$$

Provided that the Jacobian matrix $\mathbf{J}_t$ is non-singular,

$$\det \mathbf{J}_t \neq 0,$$

this system of equations can be uniquely solved with respect to $\partial g/\partial t_W$ and $\partial g/\partial t_H$. Hence, we obtain:

$$\frac{\partial g}{\partial t_W} = a(w_W, w_H, Y, n)$$
$$\frac{\partial g}{\partial t_H} = b(w_W, w_H, Y, n)$$

where $a(w_W, w_H, Y, n)$ and $b(w_W, w_H, Y, n)$ are identified functions. To interpret this rank condition, recall that $t^*_W = \zeta_W(\omega^*_W, \omega^*_H, n)$ and $t^*_H = \zeta_H(\omega^*_W, \omega^*_H, n)$. By the chain rule,

$$\mathbf{J}_t = \mathbf{J}_\omega \cdot \mathbf{J}_\zeta$$

where

$$\mathbf{J}_\omega = \begin{pmatrix} \dfrac{\partial \omega^*_W}{\partial w_W} & \dfrac{\partial \omega^*_H}{\partial w_W} \\ \dfrac{\partial \omega^*_W}{\partial w_H} & \dfrac{\partial \omega^*_H}{\partial w_H} \end{pmatrix}, \qquad \mathbf{J}_\zeta = \begin{pmatrix} \dfrac{\partial \zeta_W}{\partial \omega^*_W} & \dfrac{\partial \zeta_H}{\partial \omega^*_W} \\ \dfrac{\partial \zeta_W}{\partial \omega^*_H} & \dfrac{\partial \zeta_H}{\partial \omega^*_H} \end{pmatrix}.$$

Strict convexity of $g$ implies that $\mathbf{J}_\zeta$ is non-singular. Consequently, $\det \mathbf{J}_\omega \neq 0$ implies that $\det \mathbf{J}_t \neq 0$. We make the following, rather weak assumption.

**Invertibility condition.** The matrix

$$\mathbf{J}_\omega = \begin{pmatrix} \dfrac{\partial \omega^*_W}{\partial w_W} & \dfrac{\partial \omega^*_H}{\partial w_W} \\ \dfrac{\partial \omega^*_W}{\partial w_H} & \dfrac{\partial \omega^*_H}{\partial w_H} \end{pmatrix}$$

is non-singular.

By local invertibility of the mapping from $(w_H, w_W)$ into $(t^*_W, t^*_H)$, conditional on $(Y, n)$, the identified functions $a(w_W, w_H, Y, n)$ and $b(w_W, w_H, Y, n)$ can be expressed as functions of $(t_W, t_H, n)$. Hence, over the corresponding region of the support of $(t_W, t_H)$, the gradient $\nabla g(t_W, t_H, n)$ is identified. Since $g$ is known to depend on $(t_W, t_H, n)$ only, integration identifies $g(t_W, t_H, n)$ up

to an additive function of $n$. The remaining function is pinned down by the boundary condition in (A.18). The prices of childcare time are then given by

$$\theta_W w_W = -\frac{\partial g}{\partial t_W}\left(t_W^*, t_H^*, n\right), \qquad \theta_H w_H = -\frac{\partial g}{\partial t_H}\left(t_W^*, t_H^*, n\right)$$

from the first-order conditions of the cost minimization problem.

**Part 2.** Suppose now that $(w_W, w_H, Y, n)$ is such that $m^*(w_W, w_H, Y, n) = 0$. In that case, the non-negativity constraint on external childcare expenditure is binding, and the optimal childcare times satisfy the boundary condition:

$$g(t_W^*(w_W, w_H, Y, n), t_H^*(w_W, w_H, Y, n), n) = 0. \tag{A.22}$$

We make the following assumption.

**Support condition.** Every relevant point of the boundary $g(t_W, t_H, n) = 0$ lies in the closure of the set of optimal parental childcare time allocations $(t_W^*, t_H^*)$ associated with $m^* > 0$.

Here, the boundary refers to the boundary of the childcare technology in the $(t_W, t_H)$ space, not to the boundary between the regimes $m^* > 0$ and $m^* = 0$ in the space of exogenous variables. Under the support condition, any relevant point on $g(t_W, t_H, n) = 0$ can be approached by a sequence of optimal childcare allocations observed when $m^* > 0$. Since $g$ is identified at these points, continuity identifies $g$ at the limit. Under continuous differentiability, the same argument applies to its gradient, so that $\partial g/\partial t_W$ and $\partial g/\partial t_H$ are identified on the technological boundary as limits of their values in the region where $m^* > 0$.

The identification of the implicit prices of parental childcare time requires a further distinction. In the region where $m^* > 0$, the first-order conditions imply

$$\theta_W w_W = -\frac{\partial g}{\partial t_W}\left(t_W^*, t_H^*, n\right), \qquad \theta_H w_H = -\frac{\partial g}{\partial t_H}\left(t_W^*, t_H^*, n\right)$$

so that both implicit prices are exactly identified. At a boundary point between the regions

$m^* > 0$ and $m^* = 0$, continuity of the optimal choices and implicit prices extends this exact identification to the boundary. By contrast, at points lying strictly inside a region in which $m^* = 0$, the first-order conditions of Problem $\mathrm{P}_0$ imply that

$$\nu\theta_W w_W = -\frac{\partial g}{\partial t_W}\left(t_W^*, t_H^*, n\right), \qquad \nu\theta_H w_H = -\frac{\partial g}{\partial t_H}\left(t_W^*, t_H^*, n\right)$$

where $\nu \geq 1$ is a function of $(w_W, w_H, Y, n)$. Consequently, the two implicit prices are identified only up to a common positive multiplicative factor $\nu$. This implies that:

$$\frac{\theta_W w_W}{\theta_H w_H} = \frac{\partial g/\partial t_W}{\partial g/\partial t_H}.$$

In other words, only the relative implicit price is identified. □

**Proof of Proposition 3.** Consider first the individual problem $\mathrm{P}_I$. Under the separability assumption, the first order conditions with respect to $l_I$, $t_I$ and $c_I$, respectively, are:

$$\begin{aligned}
\frac{\partial U_I}{\partial L_I}(L_I^*, c_I^*, c_K) &= \lambda_I w_I, \\
\frac{\partial U_I}{\partial L_I}(L_I^*, c_I^*, c_K) \cdot \varphi_I'(t_I^*) &= \lambda_I(1-\theta_I) w_I, \\
\frac{\partial U_I}{\partial c_I}(L_I^*, c_I^*, c_K) &= \lambda_I,
\end{aligned}$$

where $\lambda_I$ is the Lagrange Multiplier of the spouse's budget constraint. Dividing the second condition by the first one gives:

$$\theta_I = 1 - \varphi_I'(t_I^*)$$

Eliminating the redundant equation, the first order conditions becomes:

$$\begin{aligned}
\frac{\partial U_I}{\partial L_I}(L_I^*, c_I^*, c_K) &= \lambda_I w_I, \\
\frac{\partial U_I}{\partial c_I}(L_I^*, c_I^*, c_K) &= \lambda_I,
\end{aligned}$$

Using $l_I = L_I - \varphi_I(t_I^*)$, the individual budget constraint can be written as:

$$\kappa_I - c_I - L_I w_I + \varphi_I(t_I^*) w_I - t_I^*(1-\theta_I) w_I \geq 0.$$

Define $\pi_I = \varphi_I(t_I^*) w_I - t_I^*(1-\theta_I) w_I$. The budget constraint then becomes:

$$c_I + L_I w_I \leq \kappa_I + \pi_I.$$

Thus, conditional on $\kappa_I$, the remaining choice of $L_I$ and $c_I$ is equivalent to a standard consumer problem with wage $w_I$ and virtual income $\kappa_I + \pi_I$. The first and third first-order conditions, together with this budget constraint, therefore determine $L_I^*$ and $c_I^*$, with $I = W$ and $H$.

The first-order conditions of Problem $\mathrm{P}_0$ are:

$$\nu \theta_W w_W + \frac{\partial g}{\partial t_W}(t_W^*, t_H^*, n) = 0, \tag{A.23}$$

$$\nu \theta_H w_H + \frac{\partial g}{\partial t_H}(t_W^*, t_H^*, n) = 0. \tag{A.24}$$

where $\nu \geq 1$ with $\nu = 1$ if $g(t_W^*, t_H^*, n) > 0$ and $\nu > 1$ if $g(t_W^*, t_H^*, n) = 0$. Substituting $\theta_I = 1 - \varphi_I'(t_I^*)$ in these equations, we obtain:

$$\nu(1 - \varphi_W'(t_W^*)) w_W + \frac{\partial g}{\partial t_W}(t_W^*, t_H^*, n) = 0, \tag{A.25}$$

$$\nu(1 - \varphi_H'(t_H^*)) w_H + \frac{\partial g}{\partial t_H}(t_W^*, t_H^*, n) = 0. \tag{A.26}$$

These two equations, using the condition on $\nu$, can be solved with respect to $t_W^*$, $t_H^*$ and $\nu$, giving $\theta_W = 1 - \varphi_W'(t_W^*)$ and $\theta_H = 1 - \varphi_H'(t_H^*)$. Hence $t_W^*$, $t_H^*$ and $\nu$, and therefore $\theta_H$ and $\theta_H$, are functions of $w_W, w_H$ and $n$ only. □

**Proof of Corollary 4.** Under Assumption A.1 and Proposition 3,

$$\theta_I(t_I) = 1 - \varphi_I'(t_I),$$

so that $\theta_I$ depends only on spouse $I$'s own childcare time. When $m^* > 0$, the first-order conditions imply

$$\theta_I(t_I^*) = -\frac{\partial g/\partial t_I}{w_I}.$$

Since the gradient of $g$ is identified in this region, observations with $m^* > 0$ identify the function $\theta_I(\cdot)$ over the corresponding support of childcare time. The support condition implies that every relevant value of $t_I$ arising on the technological boundary $g(t_W, t_H, n) = 0$ can be approached by values of $t_I$ associated with $m^* > 0$. Hence, by continuity, $\theta_I(t_I)$ is identified at every such value. At observations with $m^* = 0$, the first-order conditions then give

$$\nu\theta_I(t_I^*)w_I = -\frac{\partial g}{\partial t_I},$$

so that

$$\nu = -\frac{\partial g/\partial t_I}{\theta_I(t_I^*)w_I}.$$

Thus, $\nu$ is identified as well. The expression can be computed using either spouse and must yield the same value of $\nu$. □

# Appendix B : Additional Empirical Results

To assess the robustness of our results, we replicated the analysis using the 2017 wave of the PSID, which contains the same variables as the 2019 wave. The 2017 wave includes 9,607 households, to which we apply the same selection criteria as for the 2019 wave. Table A.1 reproduces Table 1 for 2017 and reports parents' weekly childcare time as measured in the PSID, together with the corresponding adjusted measures. By construction, the average adjusted values are the same as in the 2019 wave. The unadjusted averages differ only moderately, with fathers reporting about two fewer hours per week and mothers about 3.5 more hours. Table A.2 presents the descriptive statistics for the 2017 wave, while Table A.3 reports the estimated

parameters of the childcare time and expenditure equations. The resulting estimates of the $\theta_I$ parameters differ substantially from those obtained for 2019: $\theta_W$ exceeds 0.9, whereas $\theta_H$ is below 0.1. Taken literally, these estimates would imply that childcare is perceived predominantly as labor by mothers and predominantly as leisure by fathers. However, when extreme observations are excluded, as in our robustness analysis for the 2019 wave, the estimates obtained under specification B become very close to those for 2019. Overall, the 2017 results support the conclusion that the $\theta_I$ parameters are below one and probably below 0.5, but their sensitivity to sample selection prevents us from drawing a more definitive conclusion.

Table A.1: Parents' Weekly Childcare Time (in Hours, with and without Adjustment) and Weekly Childcare Expenditure (in Dollars) for the 2017 Wave

| | Mean | StDev | Min | 50% | Max | Pzeros |
|---|---|---|---|---|---|---|
| Father's Original PSID Childcare Time | 31.95 | 36.23 | 0.00 | 20.00 | 168.00 | 0.01 |
| Mother's Original PSID Childcare Time | 57.14 | 48.85 | 0.00 | 40.00 | 168.00 | 0.01 |
| Father's Adjusted Childcare Time | 9.66 | 21.59 | 0.00 | 1.29 | 121.21 | 0.01 |
| Mother's Adjusted Childcare Time | 18.97 | 28.68 | 0.00 | 5.27 | 109.81 | 0.01 |
| Childcare Expenditure | 100.50 | 125.29 | 0.00 | 57.69 | 769.23 | 0.34 |

Notes: Mean = average value, StDev = standard-deviation, Min = minimum value, 50% = median, Max = maximum value, Pzeros = proportion of zeros.

Table A.2: Descriptive Statistics of the Benchmark Sample for the 2017 Wave

| | Without Child | | With Children | |
|---|---|---|---|---|
| | Mean | StDev | Mean | StDev |
| Father's Adjusted Childcare Time | 0.00 | 0.00 | 9.66 | 21.59 |
| Mother's Adjusted Childcare Time | 0.00 | 0.00 | 18.97 | 28.68 |
| Childcare Expenditure | 0.00 | 0.00 | 100.50 | 125.29 |
| Number of Children | 0.00 | 0.00 | 1.59 | 0.49 |
| Dummy for Child under 3 | 0.00 | 0.00 | 0.73 | 0.44 |
| Husband's Leisure Time | 76.86 | 12.78 | 67.49 | 22.40 |
| Wife's Leisure Time | 83.09 | 12.72 | 67.85 | 27.88 |
| Husband's Hourly Wage | 28.93 | 22.92 | 30.84 | 26.21 |
| Wife's Hourly Wage | 23.17 | 14.55 | 27.75 | 23.23 |
| Full Expenditure (in thousand dollars) | 5.26 | 2.97 | 6.09 | 3.92 |
| Husband's Education in Years | 14.24 | 2.20 | 14.23 | 2.34 |
| Wife's Education in Years | 14.83 | 2.19 | 15.01 | 2.09 |
| Husband's Age in Years | 36.31 | 7.60 | 34.16 | 5.22 |
| Wife's Age in Years | 35.12 | 7.62 | 32.69 | 4.86 |

Notes: StDev = standard-deviation.

Table A.3: ML Estimates of Childcare Time and Expenditure Equations for the 2017 Wave

| | | A | | B | | C | | D | |
|---|---|---|---|---|---|---|---|---|---|
| Param. | Variable | Est | StErr | Est | StErr. | Est | StErr. | Est | StErr. |
| $\alpha_0$ | Intercept | -86.62 | 31.30 | -86.32 | 36.88 | -116.84 | 26.18 | -84.93 | 31.26 |
| $\alpha_1$ | Husband's Educ. | -0.54 | 9.79 | 0.71 | 8.56 | -0.48 | 7.91 | -0.48 | 9.76 |
| $\alpha_2$ | Wife's Educ. | 14.54 | 9.82 | 10.19 | 8.67 | 16.16 | 7.99 | 14.18 | 9.80 |
| $\alpha_3$ | Husband's Age | -5.03 | 13.29 | -2.78 | 12.06 | -6.75 | 10.74 | -5.26 | 13.26 |
| $\alpha_4$ | Wife's Age | 25.91 | 13.61 | 21.72 | 12.35 | 17.69 | 10.95 | 25.97 | 13.58 |
| $\alpha_5$ | Child under 3 | -1.74 | 8.53 | -4.82 | 7.65 | -9.78 | 6.69 | -1.66 | 8.50 |
| $\alpha_6$ | Number of Children | 28.65 | 8.29 | 16.92 | 7.42 | -12.99 | 7.12 | 28.56 | 8.27 |
| $\beta_{H,0}$ | Intercept | -3.25 | 3.36 | 1.88 | 4.29 | -4.47 | 2.75 | 13.71 | 5.64 |
| $\beta_{H,1}$ | Husband's Educ. | 0.90 | 1.26 | -0.24 | 1.24 | 0.25 | 1.06 | 1.81 | 2.12 |
| $\beta_{H,2}$ | Wife's Educ. | -0.48 | 1.26 | -0.63 | 1.27 | -0.29 | 1.07 | -1.01 | 2.13 |
| $\beta_{H,3}$ | Husband's Age | 0.35 | 1.74 | 0.29 | 1.78 | 0.18 | 1.46 | 0.01 | 2.93 |
| $\beta_{H,4}$ | Wife's Age | -1.89 | 1.76 | -1.37 | 1.82 | -1.74 | 1.48 | -3.62 | 2.96 |
| $\beta_{H,5}$ | Child under 3 | 0.13 | 1.09 | 0.29 | 1.10 | 0.32 | 0.90 | 0.46 | 1.83 |
| $\beta_{H,6}$ | Number of Children | -0.09 | 1.05 | -0.27 | 1.07 | 0.29 | 0.93 | -0.74 | 1.77 |
| $\beta_{W,0}$ | Intercept | 12.87 | 3.11 | 8.60 | 4.95 | 12.89 | 2.76 | 46.63 | 5.30 |
| $\beta_{W,1}$ | Husband's Educ. | 2.39 | 1.64 | 1.30 | 1.68 | 0.75 | 1.33 | 4.42 | 2.79 |
| $\beta_{W,2}$ | Wife's Educ. | -3.32 | 1.67 | -2.84 | 1.77 | -2.26 | 1.38 | -4.62 | 2.85 |
| $\beta_{W,3}$ | Husband's Age | -0.37 | 2.32 | -0.11 | 2.48 | 1.04 | 1.88 | -0.32 | 3.95 |
| $\beta_{W,4}$ | Wife's Age | -0.97 | 2.35 | -0.95 | 2.54 | -2.21 | 1.91 | -2.33 | 4.00 |
| $\beta_{W,5}$ | Child under 3 | 0.94 | 1.45 | 1.42 | 1.54 | 2.62 | 1.17 | 1.14 | 2.47 |
| $\beta_{W,6}$ | Number of Children | -0.82 | 1.41 | -1.27 | 1.50 | 0.10 | 1.19 | -2.04 | 2.40 |
| $\gamma_H$ | | 15.30 | 10.75 | 15.18 | 9.06 | 21.67 | 6.86 | 17.02 | 12.96 |
| $\gamma_W$ | | 26.16 | 6.93 | 32.69 | 10.47 | 27.25 | 5.99 | 34.20 | 8.98 |
| $\gamma_{W,H}$ | | -0.20 | 0.64 | 0.03 | 2.54 | -0.14 | 0.77 | -0.03 | 1.16 |
| $\theta_H$ | | 0.06 | 0.08 | 0.21 | 0.24 | 0.09 | 0.06 | 0.04 | 0.06 |
| $\theta_W$ | | 0.99 | 0.47 | 0.52 | 0.29 | 0.94 | 0.37 | 0.58 | 0.28 |

Notes: All demographic variables are standardized. A = the benchmark sample (425 observations), B = the sample without extreme values (370 observations), C = the sample with households with more than 2 children (623 observations), D = the sample with original PSID childcare time measures (425 observations).

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The authors used OpenAI's ChatGPT (GPT-5, 2025) to assist with minor language editing and stylistic improvements. The authors reviewed and approved all content generated by this tool.